



Novel antibiotics in the Era of Pediatric antimicrobial resistance (AMR)



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นายผดุงเกียรติ เข้มน้อย



16 พฤษภาคม 2569



โรงแรมมิเลียม เชียงใหม่





**PEDIATRIC
INFECTIOUS
DISEASE UNIT**
FACULTY OF MEDICINE,
CHIANG MAI UNIVERSITY

Novel Antibiotics in the Era of Pediatric Antimicrobial Resistance (AMR)

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Outlines



- Clinical case studies
- Understanding the microbiology
- Evidence-based treatment guidelines
- Novel antibiotics in pediatric AMR





Case 1



A 2-month-old male infant with VAP

- Term male newborn, GA 39+1 wks, BBW 3,335 gm, AGA
- **U/D:** TOF with PA with MAPCA with small VSD
- History of recurrent pneumonia
- **Operation:** S/P RVOT reconstruction with MAPCA ligations
- Post-op antibiotics: cefazolin 100 mg/kg/day
- Post-op day 3: high grade fever
- Sputum G/S:
 - few gram negative bacilli, coccobacilli





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- 

What is the proper management in this patient?

- A. Step up to ceftazidime
 - B. Step up to piperacillin/tazobactam
 - C. Step up to meropenem prolonged infusion
 - D. Sulbactam-based + polymyxin E
 - E. Sulbactam-based + polymyxin B
 - F. Consult ID
- 

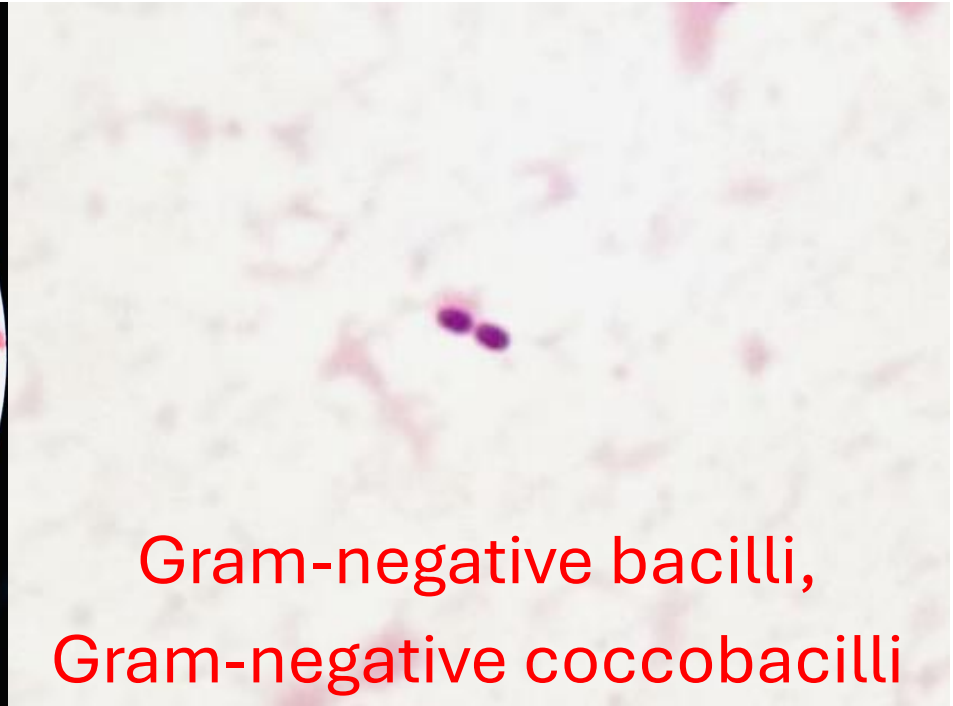
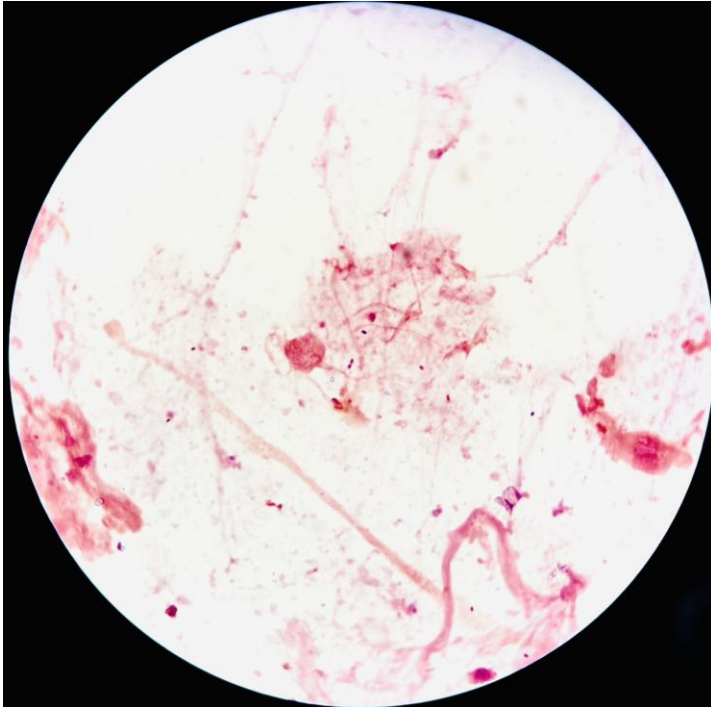


เป็นเชื้ออะไรได้บ้าง?

ขอปรึกษาห้องแลป?

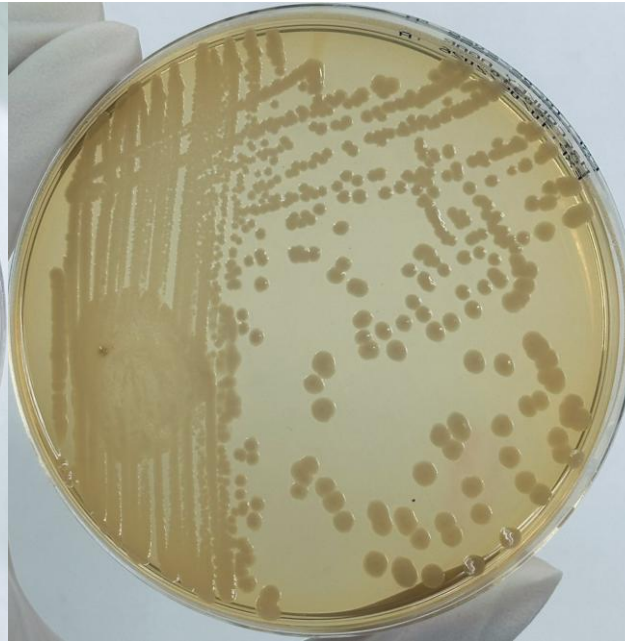


Microscopic examination: Gram's stain



Gram-negative bacilli,
Gram-negative coccobacilli

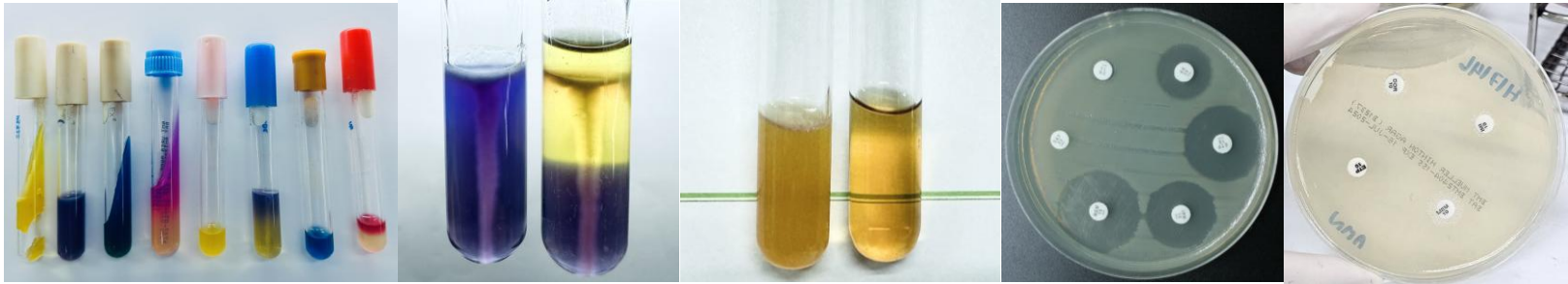
Macroscopic examination: Colony morphology



Non-lactose fermenter
(NLF)
on MacConkey agar

Identification and Antimicrobial Susceptibility Testing (AST)

- Conventional method



- Automation

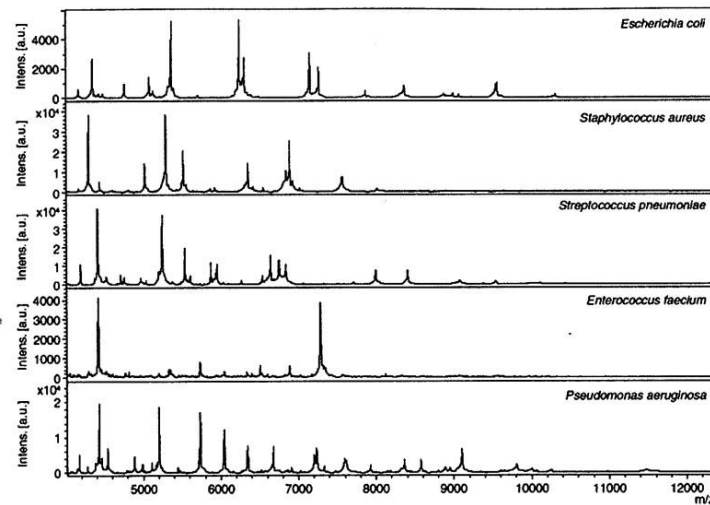
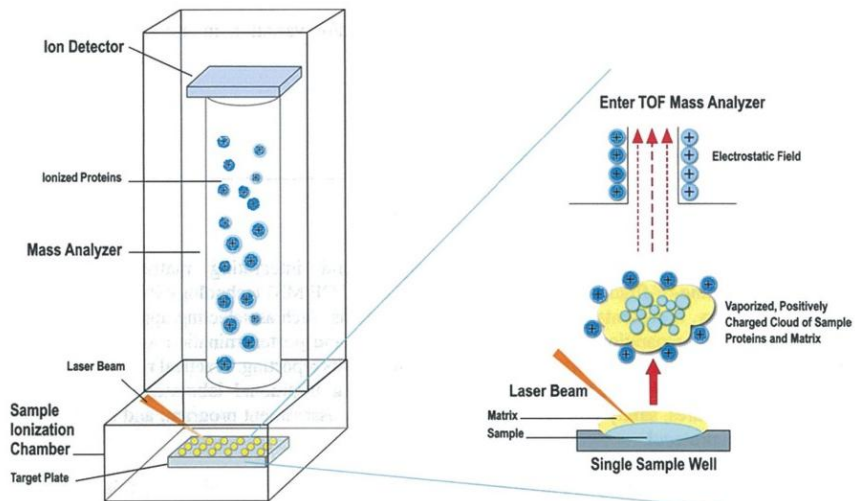


MALDI-TOF MS

Matrix-Assisted Laser
Desorption/Ionization Time-of-Flight
Mass Spectrometry



α -Cyano-4-Hydroxycinnamic
acid: CHCA



Clinical Microbiology's Guidelines



- *Enterobacterales*
- *Pseudomonas aeruginosa*
- *Acinetobacter* spp.
- *Stenotrophomonas maltophilia*
- Non-Enterobacterales
- *Haemophilus influenzae* & *H. parainfluenzae*
- *Neisseria gonorrhoeae*
- *Neisseria meningitidis*
- *Staphylococcus* spp.
- *Enterococcus* spp.
- *Streptococcus pneumoniae*
- *Streptococcus* spp. (Beta-hemolytic)
- *Streptococcus* spp. (Viridans gr.)
- Anaerobes



- *Abiotrophia*, *Granulicatella* spp. (nutritional variant *Streptococcus*)
- *Aerococcus* spp.
- *Bacillus* spp. (not *B. anthracis*)
- *Corynebacterium* spp. (including *C. diphtheriae*) and *coryneform*
- *Erysipelothrix rhusiopathiae*
- *Gemella* spp.
- *Lactobacillus* spp.
- *Lactococcus* spp.
- *Leuconostoc* spp.
- *Listeria monocytogenes*
- *Micrococcus* spp.
- *Pediococcus* spp.
- *Rothia mucilaginosa*



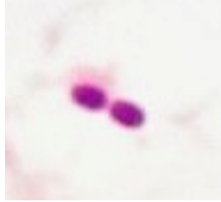
MALDI-TOF MS: *Acinetobacter baumannii*

***Acinetobacter baumannii* complex**

- *Acinetobacter baumannii*
- *Acinetobacter calcoaceticus*
- *Acinetobacter nosocomialis*
- *Acinetobacter pittii*
- *Acinetobacter seifertii*
- *Acinetobacter dijkshoorniae*



Acinetobacter baumannii



Gram-negative

Bacilli

Cocci

Obligate anaerobic

Bacteroides spp.
Prevotella spp.
Porphyromonas spp.
Fusobacterium spp.

Aerobic/Facultative anaerobic

Glucose fermenter

Oxidase positive

Non-lactose fermenter

Vibrio spp.
Aeromonas spp.
Plesiomonas spp.
Pasteurella spp.

Oxidase negative

Enterobacterales

Lactose fermenter

Escherichia coli
Klebsiella spp.
Enterobacter spp.
Citrobacter spp.
Serratia spp.

Non-lactose fermenter

Proteus spp.
Providencia spp.
Morganella spp.
Salmonella spp.
Shigella spp.
Yersinia spp.

Non-glucose fermenter

Non-lactose fermenter

Oxidase positive

Pseudomonas spp.

Oxidase negative

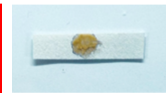
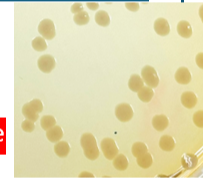
Acinetobacter spp.
S. maltophilia
B. cepacia

Obligate anaerobic

Vellonella spp.

Aerobic/Facultative anaerobic

Moraxella catarrhalis
Neisseria spp.



Identification of genus *Acinetobacter*

TABLE 1 Oxidase-negative, indole-negative, nonfermentative, Gram-negative rods: the genus *Acinetobacter*^a

Characteristic	<i>A. baumannii</i> (25) ^{b,c}		<i>A. calcoaceticus</i> (11) ^b	<i>A. lactucae</i> (6) ^b		<i>A. nosocomialis</i> (20) ^b		<i>A. pittii</i> (20) ^b		<i>A. seifertii</i> (16) ^b		<i>A. beijerinckii</i> (15) ^d		<i>A. bereziniae</i> (16)		<i>A. colsonii</i> sistens (24) ^d		<i>A. courvalinii</i> (9) ^d		<i>A. dispersus</i> (9) ^d		<i>A. gyllenbergii</i> (17)		<i>A. gyllenbergii</i> (9) ^d		<i>A. haemolyticus</i> (16) ^d		<i>A. indicus</i> (2)		<i>A. johnsonii</i> (30)		<i>A. junii</i> (14) ^d		<i>A. luoffii</i> (37)		<i>A. modestus</i> (7) ^d		<i>A. portus</i> (10) ^d		<i>A. proteolyticus</i> (6) ^d		<i>A. pseudoluoffii</i> (15)		<i>A. radioresistens</i> (12)		<i>A. schindleri</i> (22)		<i>A. soli</i> (5)		<i>A. ursingii</i> (29)		<i>A. variabilis</i> (16)		<i>A. vivianii</i> (9) ^d																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
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^aListed are only the species found in humans. All data were obtained in the Laboratory of Bacterial Genetics in Prague (A. Nemeš) and are taken from previous studies (76, 80). The numbers are percentages of strains positive in a given test.

^bNumeral in parentheses after species names are numbers of strains tested.

^cSpecies of the *A. calcoaceticus*-*A. baumannii* complex (71–73).

^dSpecies of the hemolytic clade (57).

^eUnreproducible or delayed growth of most strains.

^fWeak growth of most strains with positive reactions.

^gGrowth tested at 38°C instead of 37°C.

The preferred antibiotic options for the treatment of CRAB infections

Agent	Percentages of In Vitro Activity Against CRAB Isolates
Polymyxin B/Colistin	80.3–99.3% (Defined as colistin MICs ≤ 2 mg/L)
Tigecycline	No breakpoints established; MIC ₅₀ = 1–4 mg/L; MIC ₉₀ = 2–8 mg/L
Ampicillin-sulbactam	3.7–24.3% (Defined by CLSI ampicillin- sulbactam breakpoint of $\leq 8/4$ mg/L)
Minocycline	54–72.1% (Defined by the CLSI breakpoint of ≤ 4 mg/L)
Cefiderocol	89.7–96.1% (Defined by the CLSI breakpoint of ≤ 4 mg/L)
Sulbactam-durlobactam	96.7–96.8% (Based on FDA provisional breakpoint of ≤ 4 mg/L)

General facts about colistin and polymyxin B

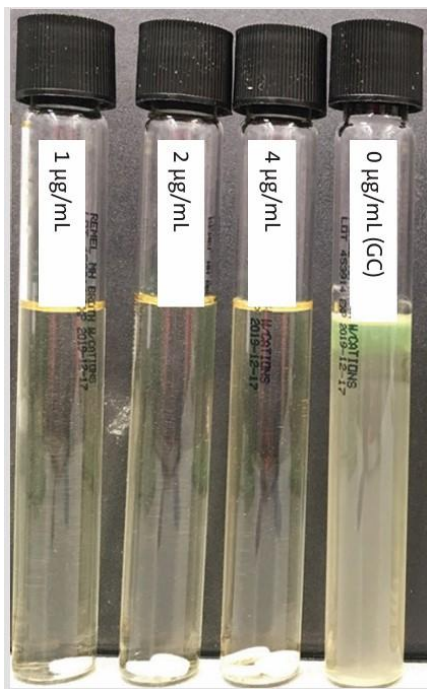
- They have **limited clinical efficacy**, even if an intermediate result is obtained.
- Alternative agents are strongly preferred.
- They should be used in **combination** of one or more active antimicrobial agents.
- For **colistin**, **broth microdilution**, colistin broth disk elution (**CBDE**) and colistin agar test (**CAT**) are acceptable.
- For **polymyxin B**, only the **broth microdilution** are acceptable.
- **Disk diffusion** and gradient diffusion methods (**E-test**) should **not be performed**.
- Colistin and polymyxin B are considered equivalent agents.
- MICs obtained from testing colistin predict MICs to polymyxin B and vice versa.
- The **CBDE** and **CAT** methods were evaluated for *Acinetobacter* spp. by CLSI and found to yield inaccurate results. Therefore, these 2 methods can be used for *Enterobacterales* and *P. aeruginosa* only, **NOT for *Acinetobacter* spp.**

Zone Diameter and MIC Breakpoints for *Acinetobacter* spp.

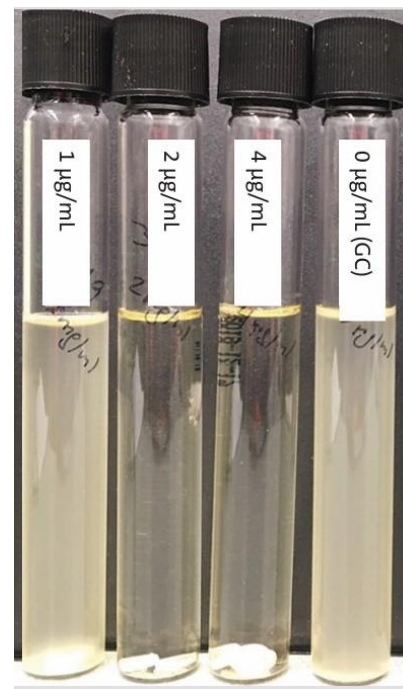
Table 2B-2. *Acinetobacter* spp. (Continued)

Antimicrobial Agent	Disk Content	Interpretive Categories and Zone Diameter Breakpoints, Nearest Whole mm			Interpretive Categories and MIC Breakpoints, µg/mL			Comments
		S	I	R	S	I	R	
LIPOPEPTIDES (Continued)								
Colistin or polymyxin B	—	—	—	—	—	≤ 2	≥ 4	(9) Colistin (methanesulfonate) should be given with a loading dose and maximum renally adjusted doses (see international consensus guidelines⁴). (10) Polymyxin B should be given with a loading dose and maximum recommended doses (see international consensus guidelines⁴). (11) When colistin or polymyxin B is given systemically, the drug is unlikely to be effective for pneumonia. (12) The only approved MIC method is broth microdilution. CBDE, CAT, disk diffusion, and gradient diffusion should not be performed. See comment (6).
	—	—	—	—	—	≤ 2	≥ 4	

Colistin Broth Disk Elution (CBDE)

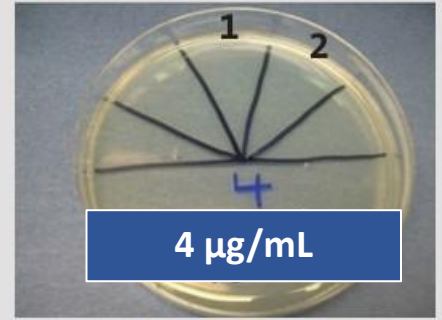
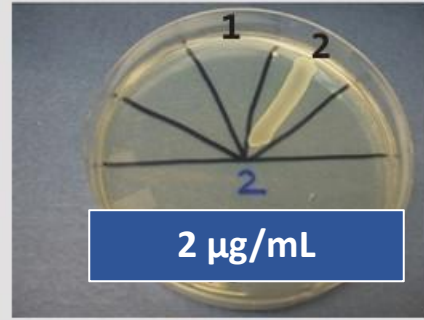
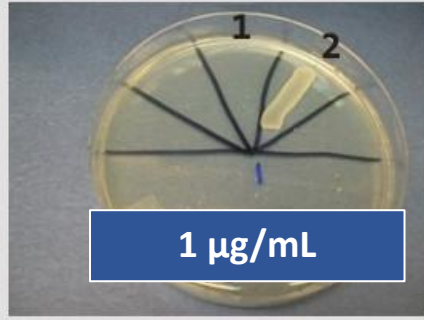
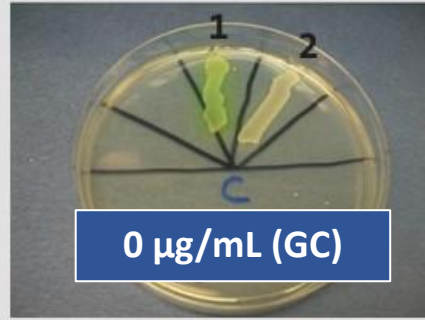


P. aeruginosa ATCC® 27853
with an MIC $\leq 1 \mu\text{g/mL}$ ($\leq 1-4 \mu\text{g/mL}$)



E. coli ATCC® BAA-3170™
with an MIC $2 \mu\text{g/mL}$ ($\leq 1-4 \mu\text{g/mL}$)

Colistin Agar Test (CAT)



Position 1

- *P. aeruginosa* ATCC® 27853 with an MIC ≤ 1 µg/mL ($\leq 1-4$ µg/mL)

Position 2

- *E. coli* ATCC® BAA-3170™

(formerly *E. coli* AR Bank #0349 mcr-1) with an MIC 4 µg/mL ($\leq 1-4$ µg/mL)

Tests for Colistin and Polymyxin B Resistance for *Enterobacterales*, *P. aeruginosa* and *Acinetobacter* spp.

Colistin	MIC			Antimicrobial susceptibility testing methods				
	S	I	R	BMD	CBDE	CAT	Disk diffusion	Gradient diffusion
<i>Enterobacterales</i>	-	≤2	≥4					
<i>P. aeruginosa</i>	-	≤2	≥4					
<i>Acinetobacter</i> spp.	-	≤2	≥4					
Polymyxin B	MIC			Antimicrobial susceptibility testing methods				
	S	I	R	BMD	CBDE	CAT	Disk diffusion	Gradient diffusion
<i>Enterobacterales</i>	-	≤2	≥4					
<i>P. aeruginosa</i>	-	≤2	≥4					
<i>Acinetobacter</i> spp.	-	≤2	≥4					

BMD: Broth microdilution
 CBDE: Colistin broth disk elution
 CAT: Colistin agar test

How to perform tigecycline MIC ?

III. Reporting Results

A. Organisms Included in Tables 2

The MIC values determined as described in CLSI M07² may be reported directly to clinicians for patient care purposes. However, it is essential that an interpretive category result (S, SDD, I, R, or NS) also be provided routinely to facilitate understanding of the MIC report by clinicians. Zone diameter measurements without an interpretive category should not be reported. Recommended interpretive categories for various MIC and zone diameter values are included in tables for each organism group and are based on the evaluation of data as described in CLSI M23.⁴

Laboratories should report results only for agents listed in Tables 2 specific to the organism being tested. It is not appropriate to apply disk diffusion or MIC breakpoints borrowed from a table in which the organism is not listed. There may be rare cases for which an agent may be appropriate for an isolate but for which there are no CLSI breakpoints (eg, tigecycline). In these cases, the FDA Susceptibility Test Interpretive Criteria website (<https://www.fda.gov/drugs/development-resources/fda-recognized-antimicrobial-susceptibility-test-interpretive-criteria>) and the prescribing information document for the agent should be consulted.

CLSI	FDA
Tier 1, Tier 2, Tier 3, Tier 4	
Inv. (investigational agent)	
	Tigecycline

FDA-Recognized Antimicrobial Susceptibility Test Interpretive Criteria

Drug	Route of Administration	STIC for Drug Included in the Recognized CLSI Standard	Exceptions or Additions to the Recognized CLSI Standard
Ticarcillin and clavulanate	Injection	Yes	Yes
Tigecycline	Injection	No	Yes
Tobramycin	Injection	Yes	Yes
Trimethoprim hydrochloride	Oral	Yes	No
Trimethoprim and sulfamethoxazole	Oral, Injection	Yes	Yes
Vancomycin	Injection	Yes	Yes
Vancomycin*	Oral	No	No

*No STIC are recognized by FDA for this drug at this time.

Tigecycline – Injection products

FDA-Identified Interpretive Criteria

Pathogen	Minimum Inhibitory Concentrations (mcg/mL)			Disk Diffusion (zone diameter in mm)		
	S	I	R	S	I	R
<i>Staphylococcus aureus</i> (including methicillin-resistant isolates)	≤0.5 ^a	-	-	≥19	-	-
<i>Streptococcus</i> spp. other than <i>S. pneumoniae</i>	≤0.25 ^a	-	-	≥19	-	-
<i>Streptococcus pneumoniae</i>	≤0.06 ^a	-	-	≥19	-	-
<i>Enterococcus faecalis</i> (vancomycin-susceptible isolates)	≤0.25 ^a	-	-	≥19	-	-
Enterobacterales ^b	≤2	4	≥8	≥19	15-18	≤14
<i>Haemophilus influenzae</i>	≤0.25 ^a	-	-	≥19	-	-
Anaerobes ^c	≤4	8	≥16	-	-	-

S = Susceptible; I = Intermediate; R = Resistant

For disk diffusion, use paper disks impregnated with 15 mcg tigecycline

^a The current absence of resistant isolates precludes defining any results other than “Susceptible.” Isolates yielding MIC results suggestive of “Nonsusceptible” category should be submitted to reference laboratory for further testing.

^b Tigecycline has decreased *in vitro* activity against *Morganella* spp., *Proteus* spp. and *Providencia* spp.

^c Agar dilution

EUCAST Clinical Breakpoint for *Acinetobacter* spp.

Tetracyclines	MIC breakpoints (mg/L)			Disk content (µg)	Zone diameter breakpoints (mm)		
	S ≤	R >	ATU		S ≥	R <	ATU
Doxycycline	-	-			-	-	
Eravacycline	IE	IE			IE	IE	
Minocycline	IE ¹	IE ¹			IE	IE	
Tetracycline	-	-			-	-	
Tigecycline	IE	IE			IE	IE	

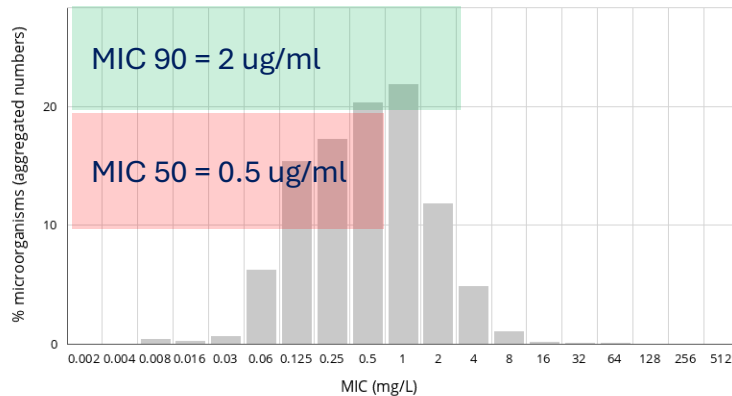
IE = insufficient evidence

Note:

- 1. Minocycline has been discussed as alternative therapy in *Acinetobacter* infections.
- The “IE” in the table pertains to intravenous therapy only. Oral administration will not accomplish sufficient exposure.

EUCAST MIC distributions for *Acinetobacter baumannii*, 2025-03-04

Tigecycline / *Acinetobacter baumannii*
International MIC distribution - Reference database 2025-03-04
Based on aggregated distributions
MIC distributions include collated data from multiple sources, geographical areas and time periods and can never be used to infer rates of resistance



MIC
Epidemiological cut-off (ECOFF): ID
Wildtype (WT) organisms: -

Confidence interval: -
34830 observations (16 data sources)

	0.002	0.004	0.008	0.016	0.03	0.06	0.125	0.25	0.5	1	2	4	8	16	32	64	128	256	512	Distributions	Observations	(T)ECOFF	Confidence interval
Tigecycline	0	0	119	59	223	2153	5370	6025	7078	7622	4106	1683	352	36	2	2	0	0	0	16	34830	ID	

ID = insufficient
data

Zone Diameter and MIC Breakpoints for *Acinetobacter* spp.

Antimicrobial Agent	Disk Content	Interpretive Categories and Zone Diameter Breakpoints, Nearest Whole mm			Interpretive Categories and MIC Breakpoints, µg/mL			Comments
		S	I	R	S	I	R	
PENICILLINS								
Piperacillin*	100 µg	≥ 21	18-20	≤ 17	≤ 16	32-64	≥ 128	
β-LACTAM COMBINATION AGENTS								
(4) Organisms that test susceptible to the β-lactam agent alone are also considered susceptible to the β-lactam combination agent. However, organisms that test susceptible to the β-lactam combination agent cannot be assumed to be susceptible to the β-lactam agent alone. Similarly, organisms that test intermediate or resistant to the β-lactam agent alone may be susceptible to the β-lactam combination agent.								
Ampicillin-sulbactam	10/10 µg	≥ 22	17-21	≤ 16	≤ 8/4	16/8	≥ 32/16	
Piperacillin-tazobactam	100/10 µg	≥ 21	18-20	≤ 17	≤ 16/4	32/4-64/4	≥ 128/4	

Carbapenem-resistant *A. baumannii* (CRAB)

- Definition of carbapenem-resistant *A. baumannii* (CRAB) **not susceptible to carbapenem** antibiotics (ie, intermediate or resistant).
- **36-45% CRAB** in the United States.
- **66-86% CRAB** in Asia and Latin America.
- CRAB isolates harbor carbapenemases, with **OXA-23** being the most common.
- **No breakpoint**-setting organizations—**neither CLSI, FDA, nor EUCAST**—provide **tigecycline** breakpoints for *Acinetobacter* spp.
- So, tigecycline and *Acinetobacter* will continue to be another clinical microbiology conundrum—**used clinically**—but given the absence of breakpoints, there are substantial technical challenges in performing MIC-based testing and its interpretation.



Back to our case



A 2-month-old male infant

#	Qty	Code	Organism
1	Moderate	CRAB	Acinetobacter baumannii complex
2	Few	M	Mixed gram negative bacilli

#	Code	Antibiotic	MIC	Result
1	AN	Amikacin	>32	R
2	SAM	Ampicillin-sulbactam	>16	R
3	FEP	Cefepime	>32	R
4	CX	Cefotaxime	>32	R
5	CZ	Ceftazidime	>32	R
6	TX	Ceftriaxone	>32	R
7	CI	Ciprofloxacin	>2	R
8	CT	Colistin	≤1	I
9	DOR	Doripenem	>16	R
10	G	Gentamicin	>8	R
11	IP	Imipenem	>16	R
12	LX	Levofloxacin	>8	R
13	ME	Meropenem	>16	R
14	TZ	Piperacillin-tazobactam	>64	R
15	ST	Trimethoprim-sulfamethoxazole	>4	R

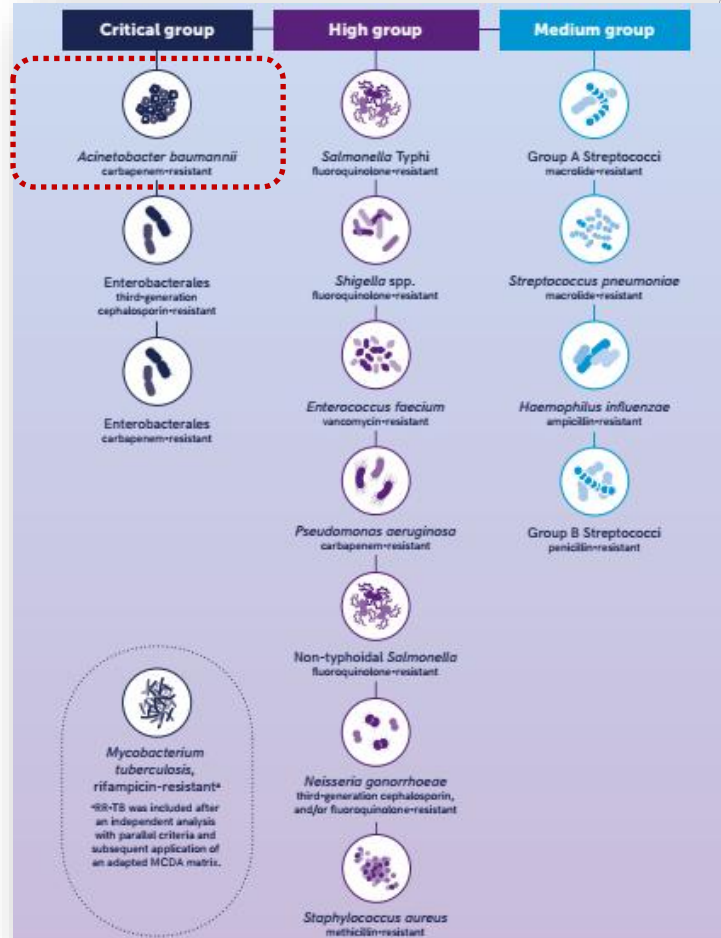
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Sputum
@ Comment:-

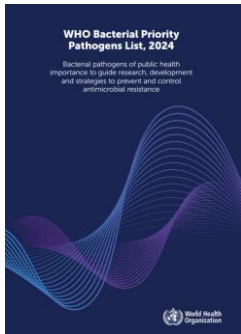
Comment



Diagnosis: VAP from Carbapenem-resistance
Acinetobacter baumannii (**CRAB**)



WHO Bacterial Priority Pathogens List, 2024 update



Percentage of antimicrobial resistance for *Acinetobacter calcoaceticus-baumannii* complex

✓ *Acinetobacter calcoaceticus-baumannii* complex



Resistance to colistin

2022: 1.7%

2023: 1.9%

Resistance to meropenem

2022: 77.5%

2023: 76.0%

Thailand One Health Dashboard on Antimicrobial Resistance

Data on monitoring and evaluation of AMR policies

ข้อมูลการติดตามและประเมินผลนโยบายการดื้อยาต้านจุลชีพของประเทศไทย

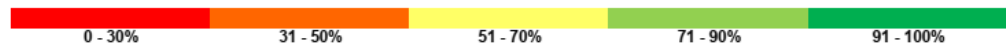
www.amrwatch.com

<https://www.thaiamrwatch.net/>. Access date: May, 11th 2026

Antibiogram CMU hospital: Jan-Dec 2024

Antimicrobial susceptible percentage (All clinical specimens)

ORGANISMS	Number of Isolates*	Trimethoprim-sulfamethoxazole	Minocycline	Colistin**	Gentamicin	Amikacin	Ampicillin-sulbactam	Amoxicillin-clavulanic acid	Piperacillin-tazobactam	Ceftazidime	Cefotaxime	Ceftazidime	Ceftazidime-avibactam	Cefepime	Imipenem	Meropenem	Doripenem	Ertapenem	Ciprofloxacin	Levofloxacin	Ampicillin
<i>Escherichia coli</i> (CRE rate = 1.48%)	2,832	52			76	97		79	91	65		72		78	99	99		99	30		
<i>Klebsiella pneumoniae</i> (CRE rate = 15.58%)	1,271	70			85	93		71	77	66		68		73	89	83		73	57		
<i>Pseudomonas aeruginosa</i> (Colistin resistance rate = 0.53%) (Carbapenem resistance rate = 99.30%)	1,133			99	90	94			85			86	91	88	84	96	88		83	84	
<i>Acinetobacter baumannii</i> complex (Colistin resistance rate = 1.17%) (Carbapenem resistance rate = 70.48%)	769	40		99	38	46	27		31	31	29	32		31	31	30	30		32	34	
<i>Proteus</i> spp.	361	59			72	96		94	100	81	78	97		99	100	100	100	100	61	62	
<i>Stenotrophomonas maltophilia</i>	323	96	86																	83	
<i>Enterobacter cloacae</i> complex (CRE rate = 7.99%)	313	77			85	97			69	57		61		79	96	96		92	67		
<i>Haemophilus influenzae</i>	183							99			100					100			97	97	79
<i>Haemophilus parainfluenzae</i>	174							99			100					100			64	66	90
<i>Citrobacter</i> spp. (CRE rate = 2.35%)	170	95			95	99		99	89	80		82		95	98	98		98	88		
<i>Morganella morganii</i> (CRE rate = 2.40%)	125	78			89	94			91	83	66	89		96	84	100	99	98	70	70	
<i>Salmonella enterica</i> (Intestinal)	101	87																	78		46
<i>Serratia marcescens</i> (CRE rate = 2.02%)	99	94			96	97			96	86	78	93		99	99	99	100	99	89	92	
<i>Klebsiella aerogenes</i> (CRE rate = 2.25%)	89	94			99	100			83	79		78		99	98	100		100	89		
<i>Aeromonas</i> spp.	83	92			98	100			92	83	82	90		96	95	96	96	95	98	97	
<i>Burkholderia cepacia</i> complex	78	100										96				88				82	
<i>Klebsiella oxytoca</i>	45	91			100	98		82	89	91		96		96	100	100		100	87		



A. baumannii complex

- R to colistin 1.17%
- R to carbapenem 70.48%

*The number of each cell represents the number of non-duplicate isolated.

**The number of each cell represents the intermediate percentage in colistin.



Treatment for CRAB



Clinical Infectious Diseases

IDSA GUIDELINES

IDSA
Infectious Diseases Society of America

hivma
the medicine association

OXFORD

Infectious Diseases Society of America 2024 Guidance on the Treatment of Antimicrobial-Resistant Gram-Negative Infections

Pranita D. Tamma,^{1,2} Emily L. Heit,³ Julie Ann Justo,⁴ Amy J. Mathers,⁵ Michael J. Sattin,⁶ and Robert A. Bonomo⁷

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Preferred

Sulbactam-durlobactam

plus

Carbapenem

- Meropenem
- Imipenem-cilastatin

Alternatives

*if sulbactam-
durlobactam is not
available*



**high-dose
ampicillin-sulbactam**

plus

at least 1 other agent

Options in Thailand:

- Sulbactam
- Ampicillin-sulbactam
- Cefoperazone-sulbactam

- Polymyxins:
Polymixin E (Colistin), polymyxin B
- Minocycline > tigecycline
- Cefiderocol



A 2-month-old male infant

Management in this patient (2024)

- H/C: no growth in 5 days
- Meropenem 60 mg/kg/day + colistin IV



New options?

Sulbactam-based agents



Polymixins



Sulbactam



**Ampicillin/
sulbactam**

**Cefoperazone
/sulbactam**

**Colistin
(Polymixin E)**

Polymixin B





Dosing for sulbactam-based agents

in neonate, infant and children for normal renal and liver function

Agents	Dosage		
	Neonates	Infant and children	Maximum
Sulbactam^a	PIDST: Neonate: 100 mg/kg/day Divided into - q 12 hr (preterm) - q 8 hr (term)	PIDST: 200 mg/kg/day IV divided into q 6-8 hr - Infused over 4 hr	Sulbactam max 12 g/day
Cefoperazone/sulbactam^a			Cefoperazone max 160 mg/kg/day or 8 g/day
Ampicillin/sulbactam^a	PIDST: Neonate: 100 mg/kg/day Divided into - q 12 hr (preterm) - q 8 hr (term) JPID: PNA ≤7 d: 100 mg/kg/dose of ampicillin IV q 8 hr, infused over 4 hr PNA 8-60 d: 75 mg/kg/dose of ampicillin IV q 6 hr, infused over 4 hr	PIDST: 200 mg/kg/day IV divided into q 6-8 hr - Infused over 4 hr JPID: 400 mg/kg/day of ampicillin IV divided q 4-6 hr, infused over 4 hr	Ampicillin max 400 mg/kg/day or 2 g/dose or 12 g/day <i>(for infant and children)</i>

^aAdjust for CrCl < 30 mL/min/1.73m²
 Abbreviations: PNA, Postnatal age



Sulbactam ออกฤทธิ์อย่างไร ?

มีข้อควรระวังในการใช้ยา
อย่างไรบ้าง

Polymixin B?

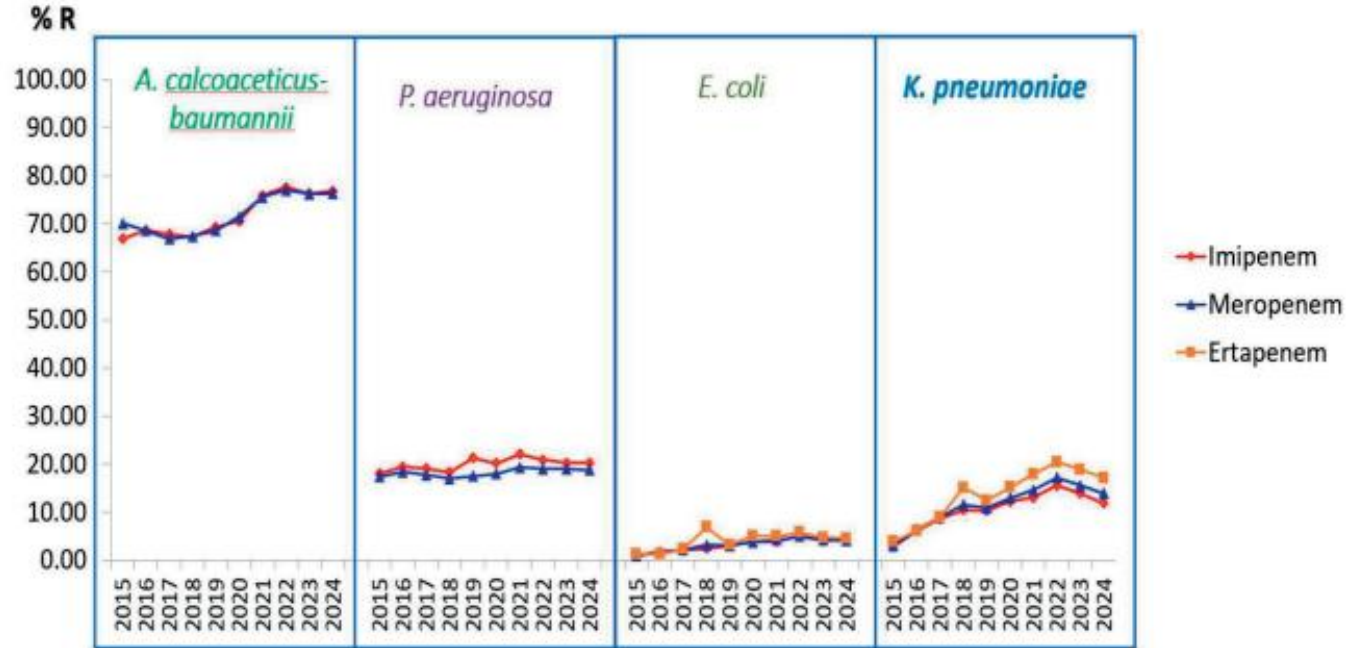




Novel drugs for treatment of antimicrobial resistance gram negative organisms

**Pattarapan Sukwuttichai, Pharm.D, BCP, FACP.
Specialized fellowship in Pediatrics Pharmacotherapy
Department of Pharmaceutical care, Faculty of Pharmacy CMU**

การดื้อยา Carbapenems ของเชื้อแกรมลบ 2015-2024



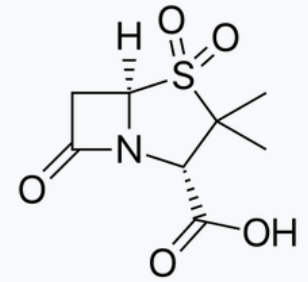
Antimicrobials against to CRAB

Combination regimens

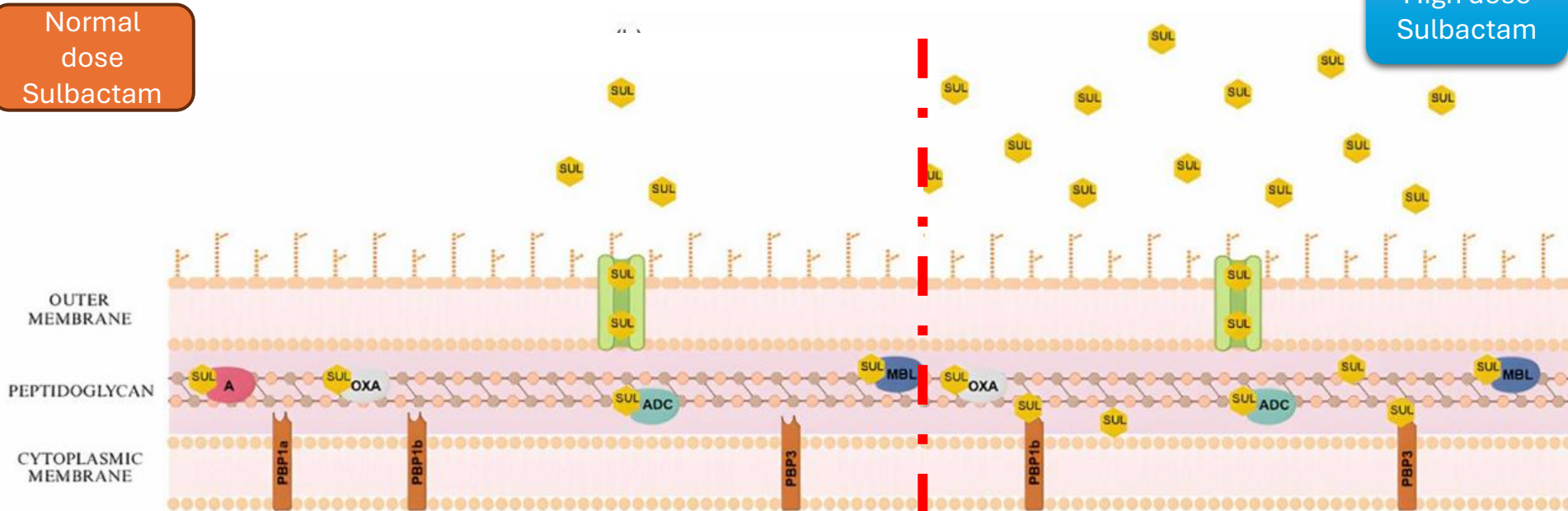
- Sulbactam/durlobactam
- Sulbactam based regimens
- Polymyxin (B, E) based regimens
- Minocyclines or Tigecycline based regimens
- Cefiderocol based regimens

Sulbactam

- Betalactase inh.
- MOA: Against PBP1a/1b, 3 of *A.baumannii*



High dose
Sulbactam



Sulbactam

- Sulbactam (mono) บัญชียา ex, ampicillin/sulbactam (ex)
cefoperazone/sulbactam (ex)
- Use: combine with Polymyxin B, E to against CRAB
- Dose
 - 200 mg/kg/day divided 3-4 doses (children)
 - Extended infusion in 4 hr (if needed)
 - 100 mg/kg/day preterm divided q 12 hr, term divided q 8 hr
- ADR: prolong prothrombin time increase risk of bleeding (from Cefoperazone-sulbactam vit K supplement 10 mg/week)

Evaluation of the efficacy of sulbactam combination therapy for monomicrobial and polymicrobial pulmonary infections caused by multidrug-resistant *Acinetobacter baumannii*

- P**: Adult patients (>18 years) with monomicrobial or polymicrobial pulmonary infections caused by MDR-AB.
- I**: Sulbactam-based treatment regimens, including cefoperazone/sulbactam and/or sulbactam sodium for injection.
- C**: Monomicrobial (mono-AB) vs. polymicrobial (poly-AB) infections.
 - Various daily dosages of sulbactam (≤ 4 g, > 6 g to < 8 g, and ≥ 8 g).
 - Different combination strategies: Sulbactam alone vs. sulbactam plus tigecycline, polymyxins, or both.
- O**: 7-day microbiological efficacy (eradication or reduction of bacterial load) and 14-day clinical efficacy (resolution or improvement of symptoms and signs).

TABLE 3 Comparison of efficacy among different doses of sulbactam^c

Sulbactam daily dosage	7-day microbiological efficacy				14-day clinical efficacy			
	Effective/total (n, %)	P	Adjusted OR (95% CI) ^a	P	Effective/total (n, %)	P	Adjusted OR (95% CI) ^b	P
≤4 g vs. ≥6 g								
≤4 g (ref)	159/268 (59.3)		(ref)		72/192(37.5)		(ref)	
≥6 g	76/98 (77.6)	0.001	2.162 (1.216–3.844)	0.009	36/89(40.4)	0.552	1.448 (0.734–2.857)	0.286
≤4 g vs. ≥6–< 8 g								
≤4 g (ref)	159/268 (59.3)		(ref)		72/192(37.5)		(ref)	
≥6–< 8 g	36/48 (75.0)	0.052	1.933 (0.929–4.025)	0.078	14/46(30.4)	0.665	0.734 (0.320–1.680)	0.464
≤4 g vs. ≥8 g								
≤4 g (ref)	159/268 (59.3)		(ref)		72/192 (37.5)		(ref)	
≥8 g	40/50 (80.0)	0.006	2.438 (1.124–5.291)	0.024	22/43 (51.2)	0.057	3.793 (1.300–11.066)	0.015
≥6–< 8 g vs. ≥8 g								
≥6–< 8 g (ref)	36/48 (75.0)		(ref)		14/46 (30.4)		(ref)	
≥8 g	40/50 (80.0)	0.632	1.261 (0.477–3.332)	0.640	22/43 (51.2)	0.035	5.170 (1.512–17.685)	0.009

^aIn the multivariate analysis of microbiological efficacy, the factors including APACHE II score, mechanical ventilation, bloodstream infection, and *P. aeruginosa* were adjusted to calculate the OR and 95% CI.

^bIn the multivariate analysis of clinical efficacy, factors such as age and urinary tract infection were adjusted to calculate the OR and 95% CI.

^cData are presented as N (%). Statistically significant *P* values are highlighted in bold.

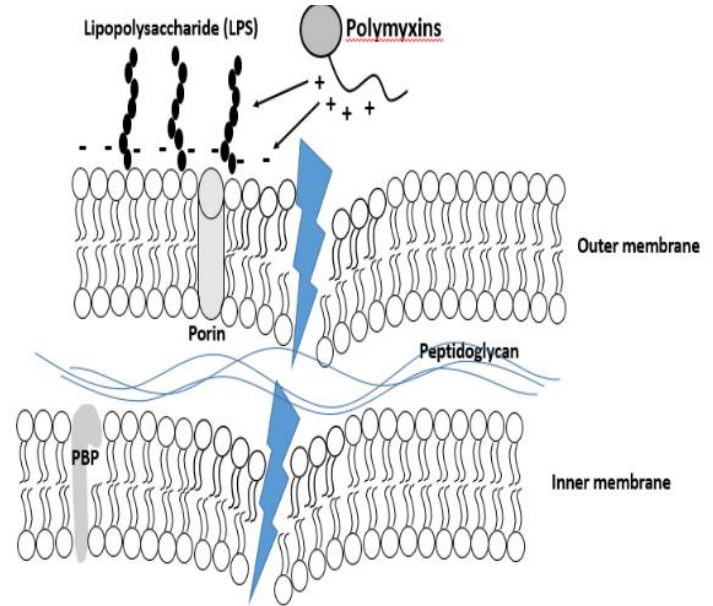
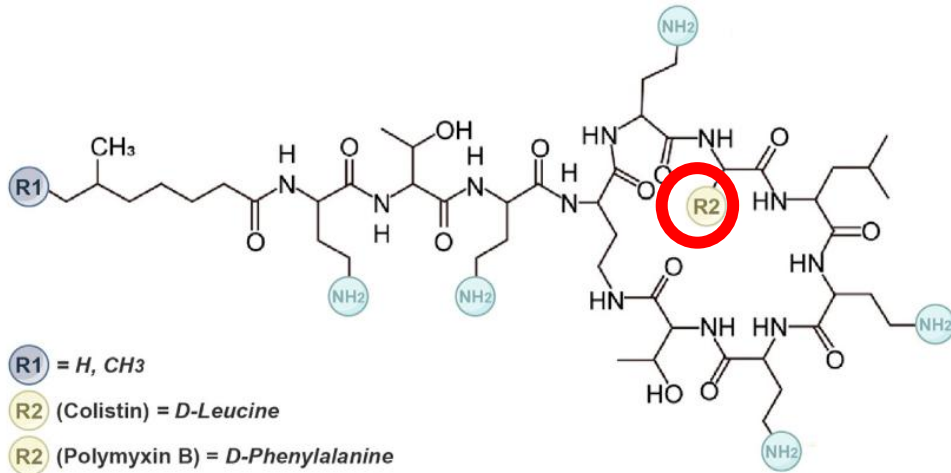
TABLE 4 Comparison of efficacy of different sulbactam combination regimens^d

Sulbactam regimen	7-day microbiological efficacy				14-day clinical efficacy			
	Effective/total (n, %)	P	Adjusted OR (95% CI) ^b	P	Effective/total (n, %)	P	Adjusted OR (95% CI) ^c	P
SUL vs. SUL + TGC + POL								
SUL-based ^a (ref)	69/153(45.1)		(ref)		41/92(44.6)		(ref)	
SUL +TGC + POL	33/36 (91.7)	< 0.001	12.256 (3.478–43.183)	< 0.001	17/48(35.4)	0.073	1.320 (0.490–3.554)	0.583
TGC vs. POL								
SUL +TGC (ref)	99/138 (71.7)		(ref)		33/109(30.3)		(ref)	
SUL +POL	34/39 (87.2)	0.059	2.850 (1.023–7.937)	0.045	17/32(53.1)	0.034	5.173 (1.471–18.191)	0.010
TGC vs. non-TGC								
SUL-based ^a (ref)	69/153 (45.1)		(ref)		41/92(44.6)		(ref)	
SUL + TGC	99/138 (71.7)	< 0.001	2.589 (1.532–4.377)	< 0.001	33/109(30.3)	0.051	0.710 (0.348–1.447)	0.346
SUL + POL (ref)	34/39 (87.2)		(ref)		17/32(53.1)		(ref)	
SUL + POL + TGC	33/36 (91.7)	0.799	1.661 (0.361–7.638)	0.515	17/48(35.4)	0.272	0.359 (0.088–1.470)	0.155
POL vs. non-POL								
SUL-based ^a (ref)	69/153 (45.1)		(ref)		41/92(44.6)		(ref)	
SUL + POL	34/39 (87.2)	< 0.001	7.379 (2.650–20.545)	< 0.001	17/32(53.1)	0.129	3.672 (1.040–12.968)	0.043
SUL + TGC (ref)	99/138 (71.7)		(ref)		33/109(30.3)		(ref)	
SUL + TGC + POL	33/36 (91.7)	0.015	4.733 (1.347–16.628)	0.015	17/48(35.4)	0.469	1.860 (0.696–4.972)	0.216

^aRefers to sulbactam alone or in combination with other antibiotics, excluding tigecycline and polymyxins.^bIn the multivariate analysis of microbiological efficacy, factors such as APACHEII score, mechanical ventilation, antibiotic course, heart disease, bloodstream infection and concurrent other pathogens were adjusted to calculate the OR and 95% CI.^cIn the multivariate analysis of clinical efficacy, factors such as age, APACHEII score, heart disease and urinary tract infection were adjusted to calculate the OR and 95%CI.^dData are presented as N (%). SUL, sulbactam; TGC, tigecycline; POL, polymyxin B or polymyxin E. Statistically significant P values are highlighted in bold.

Polymyxin B-Polymyxin E (colistin)

- Polycationic polypeptide antibiotics



รูปที่ 1. กลไกการออกฤทธิ์ของยาในกลุ่ม polymyxins

Polymyxin B-Polymyxin E (colistin)

- Coverage resistant (MDR) gram-negative bacilli, including MDR strains of *A. baumannii* (CRAB) *K. pneumoniae*, *P. aeruginosa*,
 - Not cover to *Serratia spp.*, *Proteus spp.*, *Moraxella spp.*, *Burkholderia spp.*
 - Not cover Gram Positive
- Less nephrotoxicity than colistin (polymyxin E) 39.3% vs 11.8% onset at ~Day 4 (data in adult)
- Dose : colistin LD 4-5 mg/kg maintenance 2.5 mg/kg q 12 hr
- Dose : Polymyxin B 1.25-1.5 mg/kg q 12 hr (ABW)

Comparison Table: Colistin (CMS) vs. Polymyxin B

Characteristic	Colistin / CMS / Polymyxin E	Polymyxin B	Clinical Implications
Dosing form	Prodrug (CMS must be converted to colistin)	Active drug	Plasma concentrations of colistin are more variable and take longer to achieve therapeutic levels
Dosing conversions	2.7 mg CMS = 1 mg CBA* 1 million IU = 30 mg CBA	10,000 IU = 1 mg	Variations in reporting dosages, particularly for CMS, are a source of confusion
Drug clearance	CMS undergoes renal clearance	Does not undergo renal clearance	CMS dosing must be adjusted for renal dysfunction, unlike polymyxin B
Urinary tract concentrations	High (CMS converts to colistin in the urinary tract)	Low (Not recommended in UTI)	CMS may be a preferred option for the treatment of urinary tract infections (UTIs)

Michael J. et al. Polymyxins, Editor(s): Jonathan Cohen, William G. Powderly, Steven M. Opal,

Infectious Diseases (Fourth Edition),2017, 1285-8.

*CBA = Colistin Base Activity ; IU = international units

การปรับขนาดยาในผู้ป่วยไตบกพร่อง สำหรับ maintenance dose

Antibiotic	CrCl >50	CrCl >25 - ≤50	CrCl 10 - ≤50	CrCl <10	Intermittent hemodialysis	CRRT based on eGFR >50 mL/min
Colistin (polymyxin E) (อ้างอิงข้อมูลผู้ใหญ่)	Maintenance dose 5 mg CBA/kg/day divided q12h	3.5 mg CBA/kg/day divided q12h	2.5 mg CBA/kg/day divided q12h	1.5 mg CBA/kg/day divided q12h	วันที่ไม่ได้ล้างไต 1.5 mg CBA/kg/day divided q12h วันที่ล้างไต - 1.5 mg CBA/kg/dose before dialysis - 1.5 mg CBA/kg/dose after dialysis	6 mg CBA/kg/day divided q12h

กรณี severe sepsis ในวันแรกพิจารณาให้ยาโดยไม่ต้องปรับขนาดยาตามการทำงานของไต



Case 2





A 12-year-old male with febrile neutropenia

- U/D: aplastic anemia (first diagnosis)
- Bilateral subhyaloid hemorrhage
- **Prolonged febrile neutropenia (ANC = 10)**
- **Medications:**
 - Meropenem 120 mg/kg/day
 - Vancomycin 40 mg/kg/day
 - Liposomal amphotericin B 5 mg/kg/day
- **H/C from peripheral line:**
 - **Gram negative bacilli *1 specimens**

What is the proper management in this patient?

- A. Meropenem prolonged drip in 3 hr
- B. Add colistin
- C. Ceftazidime/avibactam
- D. Ceftazidime/ avibactam + aztreonam
- E. Avibactam/ aztreonam
- F. Consult ID





True pathogen?

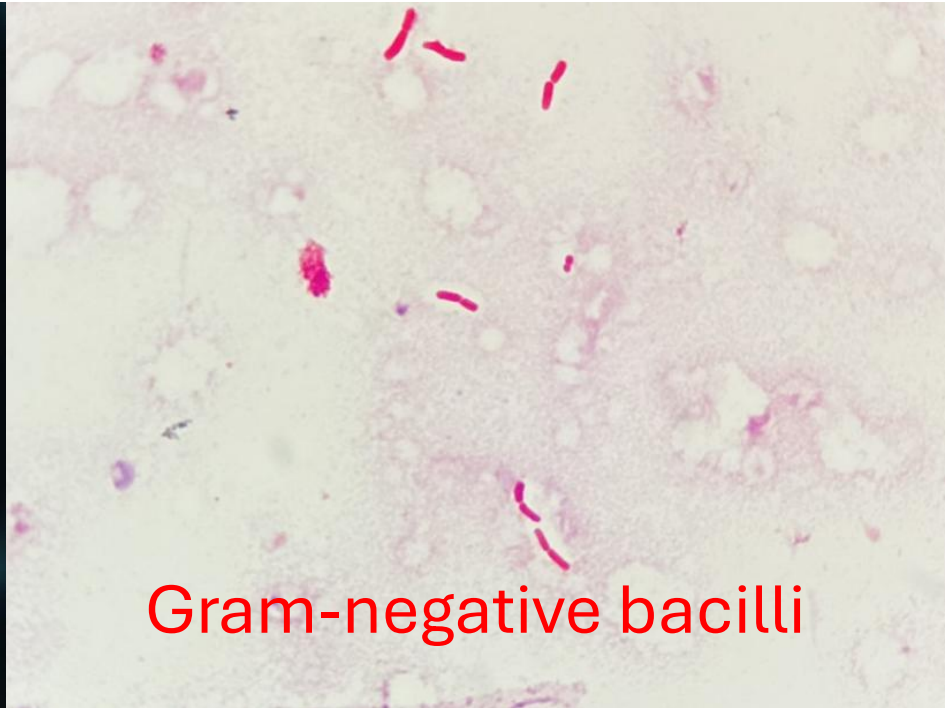
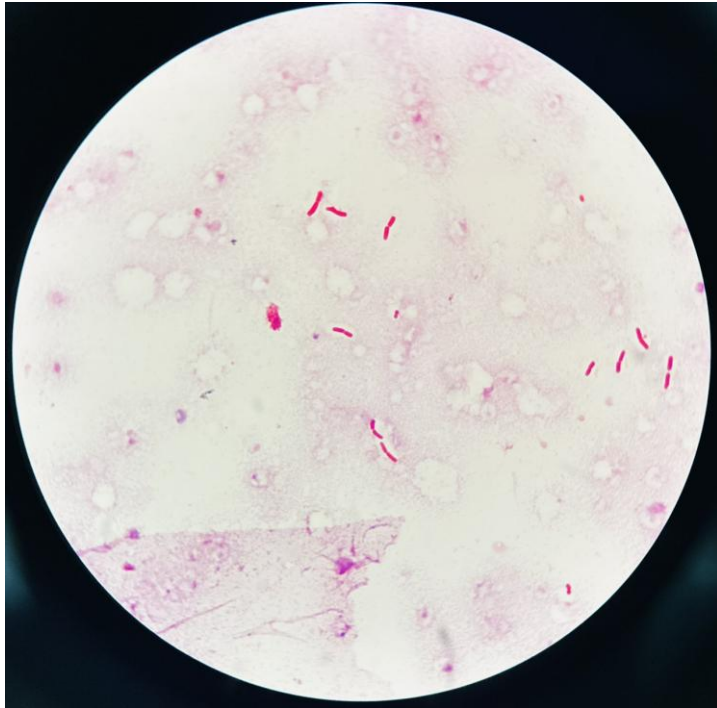


Gram negative bacilli ยังขึ้นในเลือด ขณะได้ยา
meropenem เป็นเชื้ออะไรได้บ้าง?

ต้องโทรถามอะไรเพิ่มจาก
ห้องแลปบ้าง?

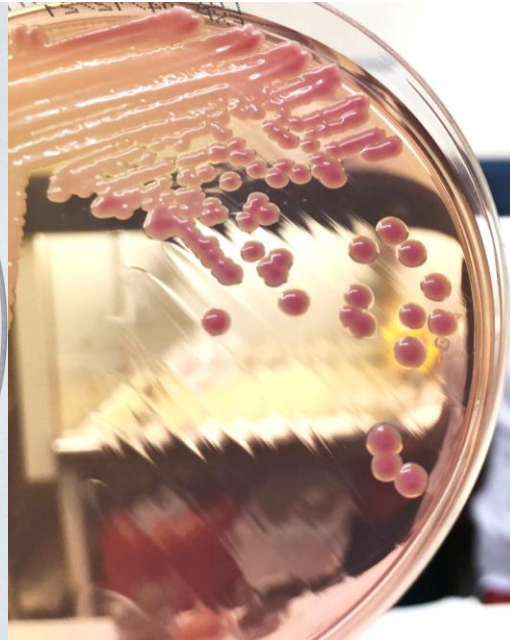


Microscopic examination: Gram's stain



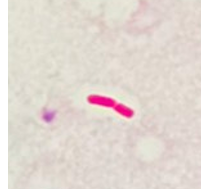
Gram-negative bacilli

Macroscopic examination: Colony morphology



Mucoid
lactose fermenter
(LF)
on MacConkey agar

Klebsiella pneumoniae



Gram-negative

Bacilli

Cocci

Obligate anaerobic

Bacteroides spp.
Prevotella spp.
Porphyromonas spp.
Fusobacterium spp.

Aerobic/Facultative anaerobic

Obligate anaerobic

Vellonella spp.

Aerobic/Facultative anaerobic

Moraxella catarrhalis
Neisseria spp.

Glucose fermenter

Non-glucose fermenter

Non-lactose fermenter

Oxidase positive

Oxidase negative

Oxidase positive

Oxidase negative

Non-lactose fermenter

Vibrio spp.
Aeromonas spp.
Plesiomonas spp.
Pasteurella spp.

Enterobacterales

Pseudomonas spp.

Acinetobacter spp.
S. maltophilia
B. cepacia

Lactose fermenter

Non-lactose fermenter

Escherichia coli
Klebsiella spp.
Enterobacter spp.
Citrobacter spp.
Serratia spp.

Proteus spp.
Providencia spp.
Morganella spp.
Salmonella spp.
Shigella spp.
Yersinia spp.



K. pneumoniae



Biochemical characterization of selected members of the genera *Klebsiella*

TABLE 4 Biochemical characterization of selected members of the genera *Klebsiella* and *Raoultella*^{a,b,c}

Species	Motility at 37°C	Indole	Urea hydrolysis	ODC	VP	Malonate	ONPG	Growth at:		Acid ^d from D-melezitose
								10°C	44°C	
<i>K. aerogenes</i>	+	–	–	+	+	+	+	+	–	–
<i>K. oxytoca</i>	–	+	+	–	+	+	+	+	+	V
<i>K. pneumoniae</i> subsp. <i>ozaenae</i>	–	–	V	–	–	–	V	–	NA	–
<i>K. pneumoniae</i> subsp. <i>pneumoniae</i> ^e	–	–	+	–	+	+	+	–	+	–
<i>K. pneumoniae</i> subsp. <i>rhinoscleromatis</i>	–	–	–	–	–	+	–	–	NA	–
<i>R. ornithinolytica</i>	–	+	+	+	V	+	+	+	NA	–
<i>R. planticola</i>	–	V	+	–	+	+	+	+	–	–
<i>R. terrigena</i>	–	–	–	–	+	+	+	+	–	+

^aAbbreviations and symbols: ODC, ornithine decarboxylase; VP, Voges-Proskauer; ONPG, *o*-nitrophenyl-β-D-galactopyranoside; NA, not available; +, ≥90%; V, variable (11 to 89%); –, ≤10%.

^bSee Table 2, footnote *b*.

^cData are from references 31, 86, 89, and 188.

^dSee Table 2, footnote *d*.

^eA negative adonitol fermentation reaction may be an indication that the strain is *K. variicola*, but this must be confirmed with gene sequencing (e.g., *rpoB*).

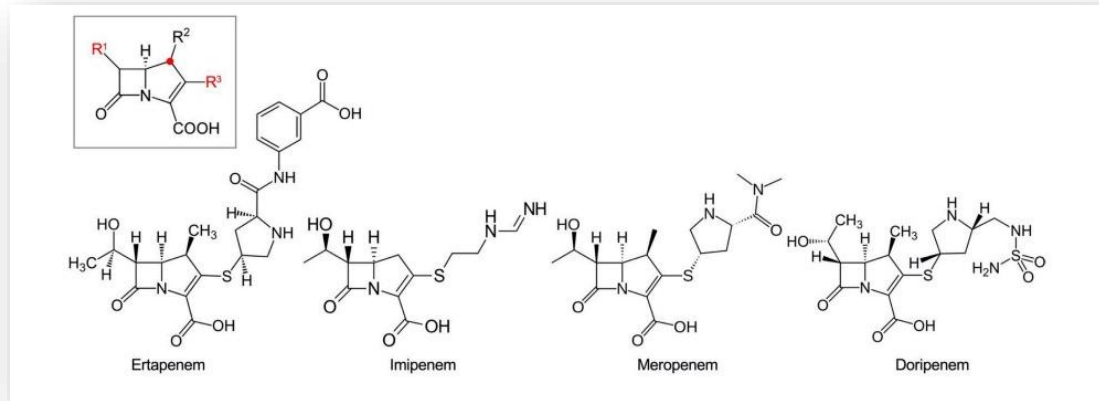
Carbapenem-resistant *Enterobacterales* (CRE)

Definition:

- *Enterobacterales* are **not susceptible** to one or more carbapenems.
- Isolates in the categories of “**intermediate**” or “**resistant**” could be called “not susceptible”

Carbapenem:

- Ertapenem
- Imipenem
- Meropenem
- Doripenem



Enterobacterales

- **Budviciaceae:** *Budvicia*, *Leminorella*, *Pragia*
- **Enterobacteriaceae:** *Klebsiella*, *Salmonella*, *Escherichia*, *Shigella*, *Citrobacter*, *Enterobacter*, *Plesiomonas*, *Cedecea*, *Kluyvera*, *Buttiauxella*, *Yokenella*, *Leclercia*, *Saccharobacter*, *Trabulsiella*, *Raoultella*, *Cronobacter*, *Mangrovibacter*, *Shimwellia*, *Gibbsiella*, *Kosakonia*, *Lelliottia*, *Pluralibacter*, *Rosenbergiella*, *Franconibacter*, *Pseudocitrobacter*, *Siccibacter*, *Apirhabdus*, *Enterobacillus*, *Phytobacter*, *Pseudescherichia*, *Izhakiella*, *Limnobaculum*, *Scandinavium*, *Huaxiibacter*, *Intestinirhabdus*, *Symbiopectobacterium*, *Tenebrionibacter*, *Tenibrionicola*
- **Gallaecimonadaceae:** *Gallaecimonas*
- **Erwiniaceae:** *Erwinia*, *Pantoea*, *Buchnera*, *Tatumella*, *Wigglesworthia*, *Phaseolibacter*, *Mixta*, *Duffyella*, *Winslowiella*
- **Hafniaceae:** *Hafnia*, *Obesumbacterium*, *Edwardsiella*
- **Morganellaceae:** *Proteus*, *Morganella*, *Providencia*, *Xenorhabdus*, *Moellerella*, *Arsenophonus*, *Photorhabdus*
- **Pectobacteriaceae:** *Pectobacterium*, *Brenneria*, *Sodalis*, *Dickeya*, *Lonsdalea*, *Acerihabitans*, *Biostraticola*, *Musicola*, *Symbiopectobacterium*
- **Yersiniaceae:** *Serratia*, *Yersinia*, *Ewingella*, *Rahnella*, *Samsonia*, *Rouxiiella*, *Chania*, *Chimaeribacter*

Carbapenem-resistant mechanisms

CP-CRE

(Carbapenemase-producing *Enterobacterales*)
Carbapenemase hydrolyzing enzymes

- **Amber Class A:**
KPC, SME, NMC, IMI, PER, GES, SFO, SFC, IBC
- **Amber Class B (MBLs):**
NDM, **IMP**, **VIM**, GIM, SIM, SPM
- **Amber Class D:**
OXA

Non-CP-CRE

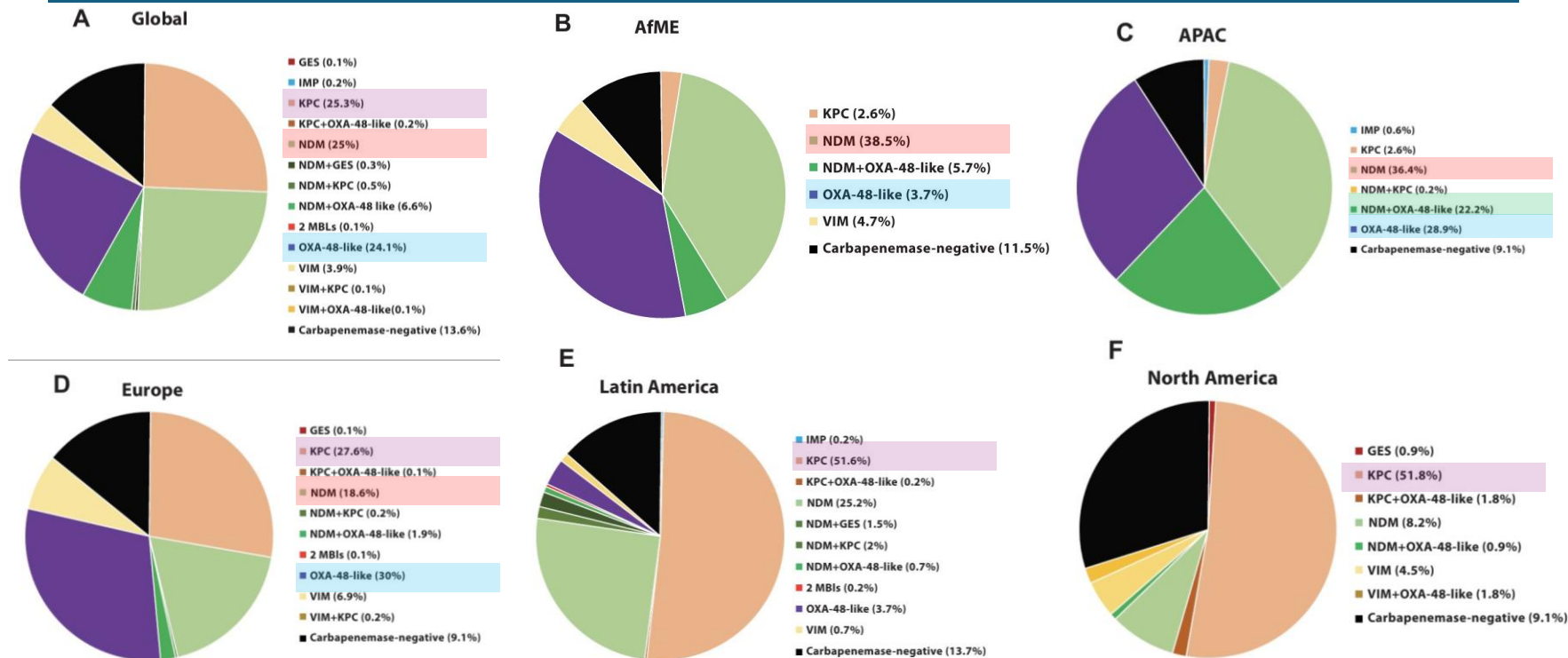
(Non-carbapenemase-producing *Enterobacterales*)

- AmpC or ESBL
- Porin mutations
- Efflux pump
- Other beta-hydrolyzing enzymes

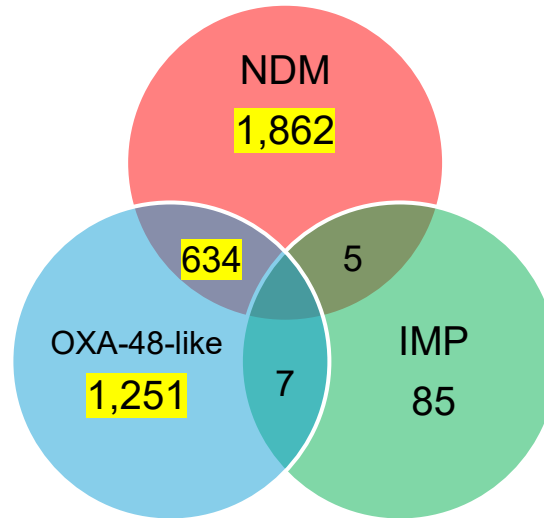
CP-CRE:

- approximately 71%-97% of CRE isolates in Thailand
- 78% of CRE isolates in Maharaj Nakorn Chiang Mai Hospital

Distribution of carbapenem resistance mechanisms



Distribution of carbapenem resistance mechanisms in Thailand



3,946 CRE isolates from 25 tertiary care hospitals during 2016-2018

Prevalence of CPKP in Maharaj Nakorn Chiang Mai Hospital

Carbapenemase-Producing *Klebsiella pneumoniae* isolates (N=117)

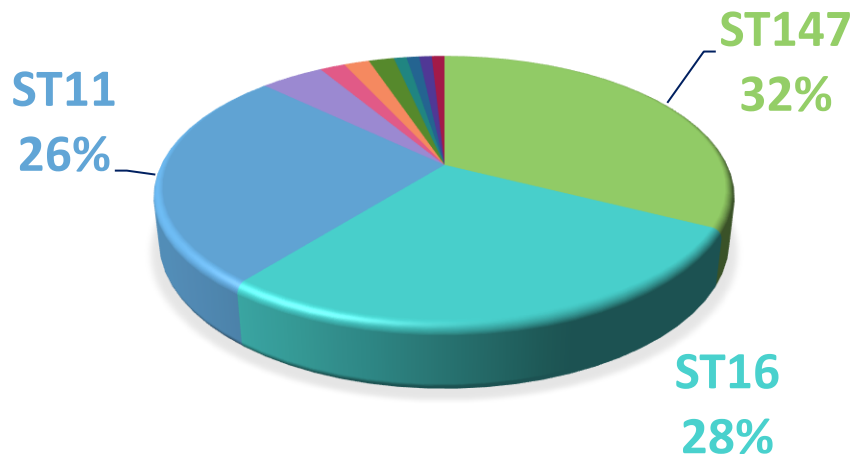


TABLE 1. DISTRIBUTION OF THE MULTILOCUS
SEQUENCE TYPING STRAINS
OF CARBAPENEMASE-PRODUCING
KLEBSIELLA PNEUMONIAE

MLST	Number of isolates (%)
ST147	38 (32.47)
ST16	33 (28.20)
ST11	31 (26.49)
ST37	5 (4.27)
ST70	2 (1.71)
ST231	2 (1.71)
ST307	2 (1.71)
ST252	1 (0.85)
ST1322	1 (0.85)
ST1441	1 (0.85)
ST2856	1 (0.85)
Total isolates	117 (100)

MLST, multilocus sequence typing; ST, sequence type.

The Distribution of Carbapenemase Genes Among Three Common Multilocus Sequence Typing Strains (N=117)

<i>MLST</i> (N)	<i>Number of isolates (%) and carbapenemase genes</i>				
	<i>bla</i> _{NDM-1}	<i>bla</i> _{OXA-181}	<i>bla</i> _{OXA-232}	<i>bla</i> _{NDM-1} + <i>bla</i> _{OXA-181}	<i>bla</i> _{NDM-1} + <i>bla</i> _{OXA-232}
ST147 (38)	34 (89.47)	2 (5.26)	—	2 (5.26)	—
ST16 (33)	8 (24.24)	—	5 (15.15)	—	20 (60.61)
ST11 (31)	—	7 (22.58)	24 (77.42)	—	—
ST37 (5)	—	—	5 (100)	—	—
ST70 (2)	1 (50)	—	1 (50)	—	—
ST231 (2)	—	—	2 (100)	—	—
ST307 (2)	—	—	2 (100)	—	—
ST252 (1)	—	—	1 (100)	—	—
ST1322 (1)	—	—	1 (100)	—	—
ST1441 (1)	1 (100)	—	—	—	—
ST2856 (1)	—	—	1 (100)	—	—
Total isolates	44 (37.29)	9 (7.63)	41 (35.04)	2 (1.69)	20 (16.95)

The Distribution of MIC₅₀ and MIC₉₀ of Carbapenemase-Producing *Klebsiella pneumoniae* Isolates (N=117)

Drug class	Antibiotics	MIC range	MIC ₅₀	MIC ₉₀
Aminoglycoside	Amikacin	≤8 to >32	≤8 ^a	16 ^b
	Gentamicin	≤2 to >8	≤2 ^a	>8
	Netilmicin	≤8 to >16	16 ^b	>16
Cephem: Cephalosporin III	Ceftriaxone	≤0.5 to >32	>32	>32
	Cefotaxime	≤1 to >32	>32	>32
	Ceftazidime	≤1 to >32	>32	>32
Cephem: Cephalosporin IV	Cefepime	≤1 to >32	>32	>32
Cephem: siderophore cephalosporin	Cefiderocol	≤0.12 to 4	0.5 ^a	1 ^a
Penem: Carbapenem	Ertapenem	2 to >4	>4	>4
	Doripenem	≤0.5 to >16	16	>16
	Imipenem	≤0.5 to >16	8	>16
β-Lactam combination	Meropenem	≤0.5 to >16	>16	>16
	Piperacillin-tazobactam	32 to >64	>64	>64
	Ceftazidime-avibactam	≤0.12 to >64	>64	>64
	Cefepime-zidebactam	≤0.12 to 2	0.5 ^a	1 ^a
	Imipenem-relebactam	≤0.12 to >64	8	>64
	Meropenem-vaborbactam	≤0.12 to >64	16	>64
Fluoroquinolone	Ciprofloxacin	≤0.06 to >2	>2	>2
	Levofloxacin	≤0.06 to >8	>8	>8
	Delaflaxacin	≤0.03 to >16	>16	>16
Folate pathway antagonist: dihydrofolate reductase inhibitor	Gatifloxacin	≤0.03 to >16	>16	>16
	Trimethoprim-sulfamethoxazole	≤1 to >4	>4	>4
Tetracycline	Tigecycline	≤0.12 to 2	0.25 ^a	1 ^a
Lipopeptide: polymyxins	Colistin	≤1 to >8	≤1 ^b	>8

^aWithin the susceptible category.
^bWithin the intermediate category.
MIC, minimum inhibitory concentration.

The Distribution of MIC₅₀ and MIC₉₀ Among Three Predominate Sequence Types of Carbapenemase-Producing *Klebsiella pneumoniae*

ST147 = **NDM**

ST16 = **NDM** + **OXA**

ST11 = **OXA**

Antibiotics	MIC range	MIC ₅₀	MIC ₉₀	MIC range	MIC ₅₀	MIC ₉₀	MIC range	MIC ₅₀	MIC ₉₀
Amikacin	≤8 to 32	16 ^a	16 ^a	≤8 to >32	16 ^a	16 ^a	≤8	≤8 ^b	≤8 ^b
Gentamicin	≤2 to >8	≤2 ^b	≤2 ^b	≤2 to >8	≤2 ^b	≤2 ^b	≤2 to >8	>8	>8
Netilmicin	≤8 to >16	>16	>16	≤8 to >16	>16	>16	≤8 to >16	≤8 ^b	>16
Ceftriaxone	1 to >32	>32	>32	>32	>32	>32	≤0.5 to >32	>32	>32
Cefotaxime	2 to >32	>32	>32	>32	>32	>32	≤1 to >32	>32	>32
Ceftazidime	≤1 to >32	>32	>32	32 to >32	>32	>32	≤1 to >32	>32	>32
Cefepime	≤1 to >32	>32	>32	>32	>32	>32	≤1 to >32	>32	>32
Cefiderocol	≤0.12 to 4	1 ^b	1 ^b	≤0.12 to 4	1 ^b	1 ^b	≤0.12 to 4	1 ^b	1 ^b
Ertapenem	4 to >4	>4	>4	4 to >4	>4	>4	2 to >4	>4	>4
Doripenem	1 to >16	16	>16	2 to >16	>16	>16	≤0.5 to >16	4	16
Imipenem	1 to >16	16	>16	1 to >16	>16	>16	≤0.5 to >16	2 ^a	8
Meropenem	2 to >16	>16	>16	8 to >16	>16	>16	1 to >16	8	>16
Piperacillin-tazobactam	>64	>64	>64	>64	>64	>64	>64	>64	>64
Ceftazidime-avibactam	0.5 to >64	>64	>64	0.5 to >64	>64	>64	≤0.12 to >64	0.5 ^b	2 ^b
Cefepime-zidebactam	≤0.12 to 2	0.25 ^b	2 ^b	0.25 to 2	1 ^b	1 ^b	≤0.12 to 2	0.5 ^b	2 ^b
Imipenem-relebactam	≤0.12 to >64	16	>64	0.25 to >64	32	>64	≤0.12 to 64	2 ^a	4
Meropenem-vaborbactam	≤0.12 to >64	16	>64	4 to >64	64	>64	0.25 to 64	8 ^a	16
Ciprofloxacin	>2	>2	>2	>2	>2	>2	2 to >2	>2	>2
Levofloxacin	4 to >8	>8	>8	8 to >8	>8	>8	2 to >8	>8	>8
Delafoxacin	2 to >16	>16	>16	1 to >16	2	8	2 to >16	>16	>16
Gatifloxacin	4 to >16	>16	>16	8 to >16	16	>16	2 to >16	>16	>16
Trimethoprim-sulfamethoxazole	≤1 to >4	>4	>4	≤1 to >4	>4	>4	≤1 to >4	>4	>4
Tigecycline	≤0.12 to 2	0.25 ^b	1 ^b	≤0.12 to 2	0.5 ^b	1 ^b	≤0.12 to 1	0.25 ^b	0.5 ^b
Colistin	≤1 to >8	2 ^a	>8	≤1 to >8	>8	>8	≤1 to >8	2 ^a	>8

^aWithin the intermediate category.

^bWithin the susceptible category.

Agents	Carbapenemase gene		
	KPC	OXA-48	NDM
Ceftazidime-avibactam			
Ceftolozane-tazobactam			
Polymyxin B / Colistin			
Tigecycline			
Aminoglycosides			
Fosfomycin			

Genotypic detection of carbapenemase gene



- **Xpert® Carba-R:** Rapid detection and differentiation of the *bla*_{KPC}, *bla*_{NDM}, *bla*_{VIM}, *bla*_{OXA-48}, and *bla*_{IMP} gene sequences from pure colonies helps clinicians optimize patient management and direct therapeutic strategy, while testing rectal and perirectal swabs helps IPs design effective infection control measures.



BIOFIRE® FILMARRAY®
PNEUMONIA PLUS (PNplus) PANEL

1 Test. 34 Targets. ~1 Hour.

ANTIMICROBIAL RESISTANCE GENES

Carbapenemases

IMP
KPC
NDM
OXA-48-like
VIM

ESBL
CTX-M

Methicillin Resistance
mecA/C and MREJ (MRSA)



BIOFIRE®
**BLOOD CULTURE IDENTIFICATION 2
(BCID2) PANEL**

1 Test. 43 Targets. ~1 Hour.

ANTIMICROBIAL RESISTANCE GENES

Carbapenemases

IMP
KPC
OXA-48-like
NDM
VIM

Colistin Resistance
mcr-1

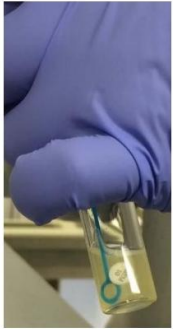
ESBL
CTX-M

Methicillin Resistance
mecA/C
mecA/C and MREJ (MRSA)

Vancomycin Resistance
vanA/B

Modified carbapenem inactivation method (mCIM)

- Phenotypic testing of carbapenemase production by **mCIM**



1 loop (1 μ l) of suspected colony

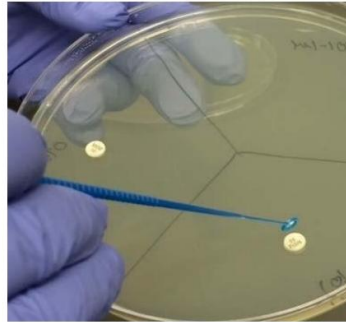


2 ml of Tryptic soy broth (TSB)



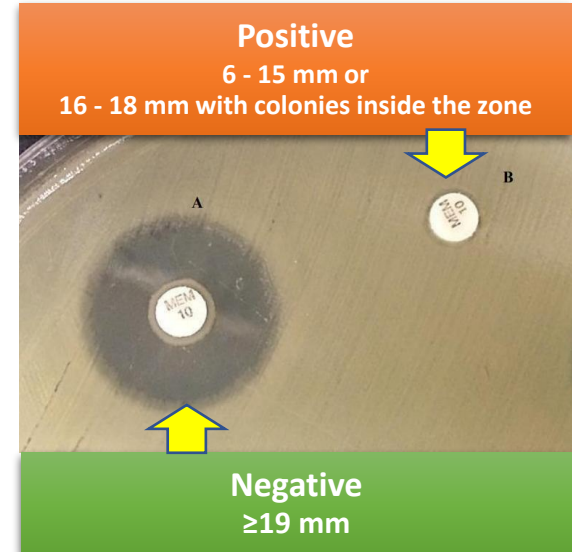
1 disc of meropenem (10 μ g)

Incubated at 35 ± 2 °C in ambient air for 4 hrs (± 15 mins)



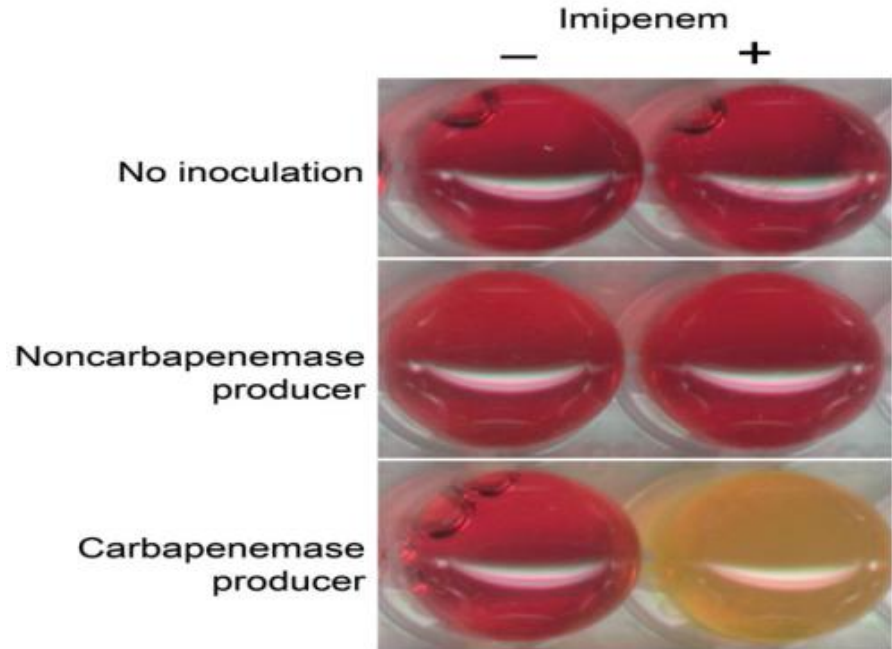
just before for 4 hrs (± 15 mins)

Spread MHA with *E. coli* ATCC 25922



Direct Methods for Detection of Carbapenemase Activity

- Colorimetric assays;
The Carba NP test
- Based on in vitro **hydrolysis of a carbapenem, imipenem**.
- **100% sensitive and specific** compared with molecular-based techniques.



The efficacy of mCIM and Carba NP test on detection of carbapenemase genes (N223)

Carbapenemase gene	mCIM Positive	Carba NP Positive
<i>bla</i> _{OXA-48} (10)	10	10
<i>bla</i> _{OXA-181} (8)	8	1
<i>bla</i> _{OXA-232} (38)	38	11
<i>bla</i> _{IMP-14} (7)	7	7
<i>bla</i> _{NDM-1} (97)	97	97
<i>bla</i> _{NDM-4} (2)	2	2
<i>bla</i> _{NDM-5} (4)	4	4
<i>bla</i> _{NDM-9} (1)	1	1
<i>bla</i> _{NDM-1} and <i>bla</i> _{OXA-181} (3)	3	3
<i>bla</i> _{NDM-1} and <i>bla</i> _{OXA-232} (53)	53	53
Total (223)	223	189

Carba NP Test for Suspected Carbapenemase Production

Table 3B, Footnote, NOTE 1:

- Test recommendations were largely derived following testing of US isolates of Enterobacterales and *P. aeruginosa* and provide for a high level of sensitivity (> 90%) and specificity (> 90%) in detecting KPC, NDM, VIM, IMP, SPM, and SME-type carbapenemases in these isolates.^{7,8} The sensitivity and specificity of the test for detecting other carbapenemase production can vary. **The ability of this test, as listed in the above procedure, to detect OXA-48-like producers is poor.**

Zone Diameter and MIC Breakpoints for Enterobacterales (excluding *Salmonella* and *Shigella* spp.)

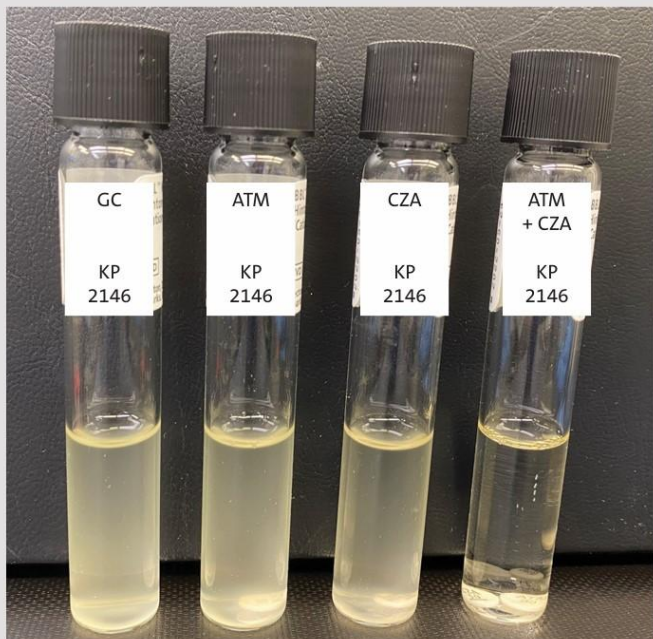
Table 2A-1. Enterobacterales (excluding *Salmonella* and *Shigella* spp.) (Continued)

Antimicrobial Agent	Disk Content	Interpretive Categories and Zone Diameter Breakpoints, Nearest Whole mm				Interpretive Categories and MIC Breakpoints, µg/mL				Comments
		S	SDD	I	R	S	SDD	I	R	
β-LACTAM COMBINATION AGENTS (Continued)										
Aztreonam-avibactam	30/20 µg	≥ 25	—	22-24^	≤ 21	≤ 4/4	—	8/4^	≥ 16/4	(11) Confirmatory MIC testing is indicated for isolates with zones of 22-24 mm to avoid overcalling susceptible results as intermediate.
Ceftazidime-avibactam	30/20 µg	≥ 21	—	—	≤ 20	≤ 8/4	—	—	≥ 16/4	(12) Confirmatory MIC testing is indicated for isolates with zones of 20-22 mm to avoid reporting false-susceptible or false-resistant results.
Ceftibuten (U, Inv.) ^a	30 µg	≥ 21	—	18-20^	≤ 17	≤ 8	—	16^	≥ 32	
MONOBACTAMS										
Aztreonam	30 µg	≥ 21	—	18-20^	≤ 17	≤ 4	—	8^	≥ 16	See comment (15).

Aztreonam Plus Ceftazidime-Avibactam Broth Disk Elution Method

Test	Aztreonam Plus Ceftazidime-Avibactam Broth Disk Elution
Organism group	Enterobacterales and <i>Stenotrophomonas maltophilia</i>
When to perform this test	Testing multidrug-resistant isolates, especially MBL producers
Test method	Tube dilution using aztreonam and ceftazidime-avibactam disks as the antimicrobial source
Medium	CAMB (5-mL tubes)
Antimicrobial concentration	30-ug aztreonam disks 30/20-ug ceftazidime-avibactam disks Final concentration: 6 ug/mL aztreonam, 6 ug/mL ceftazidime, 4 ug/mL avibactam
Inoculum	1. Using a loop or swab, pick 3-5 colonies from a fresh (18-24 h) nonselective agar plate and transfer to sterile saline (4-5 mL). 2. Adjust turbidity to equivalent of a 0.5 McFarland turbidity standard

Aztreonam Plus Ceftazidime-Avibactam Broth Disk Elution Method



- Results for routine QC strain KP 2146 demonstrating growth in the GC, ATM, and CZA tubes and no growth in the ATM + CZA tube.
- The result would be interpreted as not susceptible to ATM or CZA, but susceptible to ATM + CZA.

GC: growth control

ATM: aztreonam

CZA: ceftazidime-avibactam

ATM + CZA: aztreonam plus ceftazidime-avibactam

KP 2146: *Klebsiella pneumoniae* ATCC® BAA-2146™

Summary of evidence about treatment-emergent resistance to new anti-Gram negatives antibiotics among *Enterobacterales*

Antibiotic	Year of first approval	Year of first report of treatment-emergent resistance	Number of reported cases	Geographical distribution	Type of involved CRE	Mechanism of resistance	Cross-resistance	Resistance during combination therapy or monotherapy
CAZ/AVI	2015	2017	248	Globally reported	Most common: KPC Rare: ampC, OXA-48	Most common: KPC-2 and KPC-3 variants KPC overexpression OmpK35/OmpK36 mutations	FDC	During both mono or combo
MVB	2017	2022	2	United States and Italy	KPC	KPC overexpression OmpK-35/OmpK-36 mutations Coexpression of OXA-181 CirA mutations	CAZ/AVI IMI/REL FDC Others: aminoglycosides, fosfomycin MVB	1 case during monotherapy 1 case during combination therapy
IMI/REL	2019	2024	1	Italy	KPC	KPC overexpression OmpK-35/OmpK-36 mutations Coexpression of OXA-181 CirA mutations	Variable (CAZ/AVI, ATM/AVI)	During monotherapy
FDC	2019	2022	12	United States and Europe	Most common: NDM Rare: KPC ($n = 2$)	PBP3 mutations CMY mutations	FDC	During both mono or combo
ATM/AVI ^a	2024	2024	3 ^a	United States and Singapore	NDM-5 <i>E. coli</i>			During ceftazidime/avibactam plus aztreonam

ATM/AVI, aztreonam/avibactam; CAZ/AVI, ceftazidime/avibactam; CRE, carbapenem-resistant Enterobacterales; FDC, cefiderocol; IMI/REL, imipenem/relebactam; KPC, *Klebsiella pneumoniae* carbapenemase; MVB, meropenem/vaborbactam; NDM, New-Delhi metallo- β -lactamase.

^a In addition, in a recent study, aztreonam/avibactam resistance has been reported in 12 isolates, but no available information about *in vivo* antibiotic exposure is available (Chew et al. [20]).



Back to our case



A 12-year-old male with febrile neutropenia

Phenotypic test

Request

Bottle2#
(peripheral)

Comment

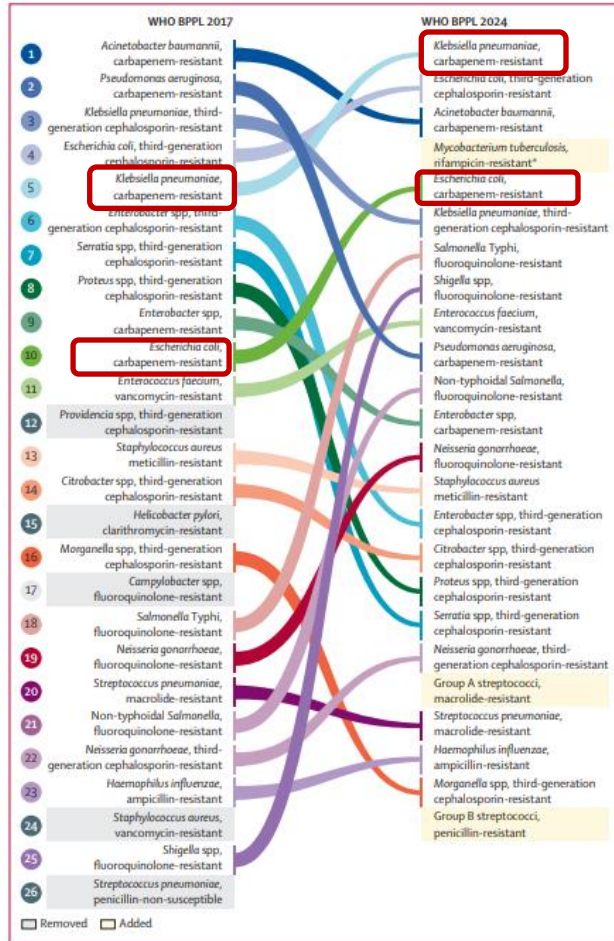
Preliminary report : gram negative bacilli
Time to positive = 14 hours. 31 mins.

#	Qty	Code	Organism	
1		CREKP	Klebsiella pneumoniae(CRE)	
#	Code	Antibiotic	MIC	Result
1	AU	Amoxicillin-clavulanic acid	>=32	R
2	TZ	Piperacillin-tazobactam	>=128	R
3	CZ	Ceftazidime	>=64	R
4	TX	Ceftriaxone	>=64	R
5	FEP	Cefepime	>=32	R
6	ETP	Ertapenem	>=8	R
7	IP	Imipenem	>=16	R
8	ME	Meropenem	>=16	R
9	AN	Amikacin	4	S
10	G	Gentamicin	>=16	R
11	CI	Ciprofloxacin	>=4	R
12	ST	Trimethoprim-sulfamethoxazole	>=16	R

R to carbapenems

Genotypic test: not done

Diagnosis: CREKP bacteremia

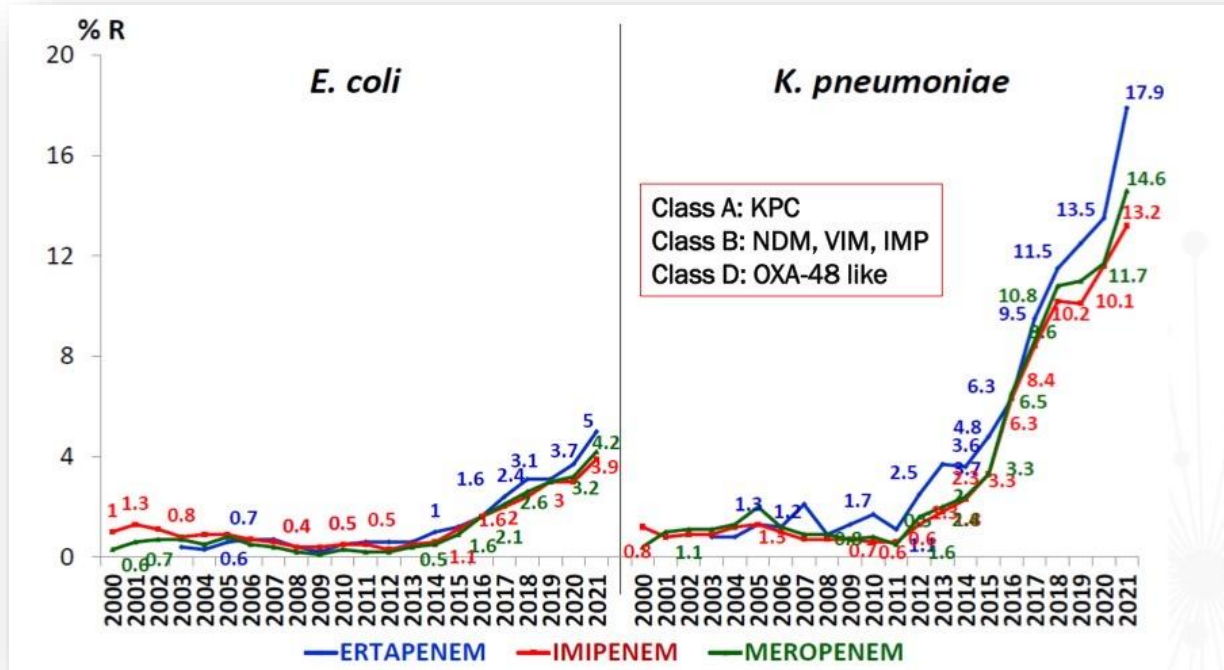


WHO Bacterial Priority Pathogens List, 2024 update

CRE; Carbapenem Resistance Enterobacterales



Carbapenem resistance in *E. coli* and *K. pneumoniae*

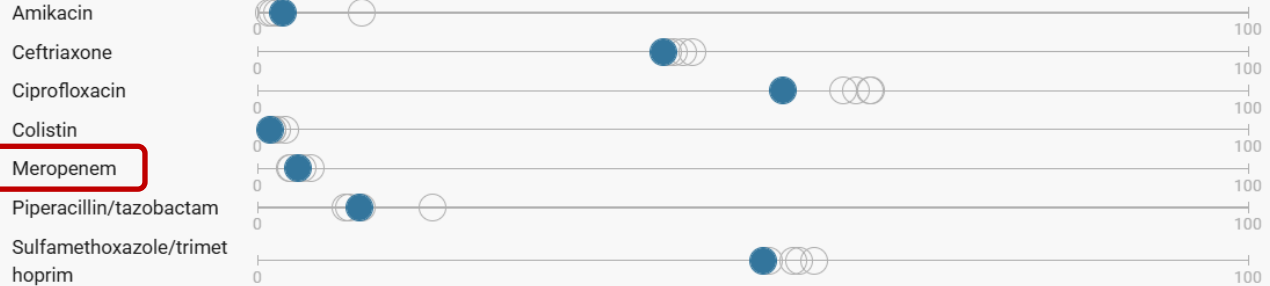




Percentage of antimicrobial resistance

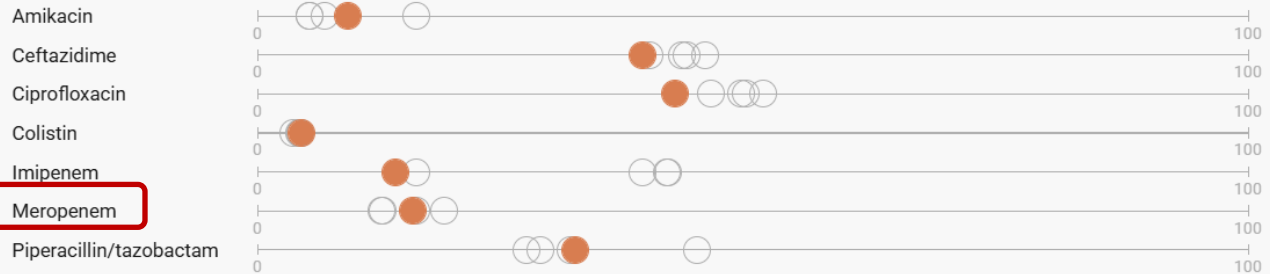
E. coli
Resistance to meropenem
2023: 4.1%

✓ *Escherichia coli*



K. pneumoniae
Resistance to meropenem
2023: 15.7%

✓ *Klebsiella pneumoniae*



Antibiogram CMU hospital: Jan-Dec 2024

Antimicrobial susceptible percentage (All clinical specimens)

CRE (*E. coli*) 1.48%

ORGANISMS	Number of Isolates*	Trimethoprim-sulfamethoxazole	Minocycline	Colistin**	Gentamicin	Amikacin	Ampicillin-sulbactam	Amoxicillin-clavulanic acid	Piperacillin-tazobactam	Ceftazidime	Cefotaxime	Ceftazidime	Ceftazidime-avibactam	Cefepime	Imipenem	Meropenem	Doripenem	Ertapenem	Ciprofloxacin	Levofloxacin	Ampicillin
<i>Escherichia coli</i> (CRE rate = 1.48%)	2,832	52			76	97		79	91	65		72		78	99	99		99	30		
<i>Klebsiella pneumoniae</i> (CRE rate = 15.58%)	1,271	70			85	93		71	77	66		68		73	89	83		73	57		
<i>Pseudomonas aeruginosa</i> (Colistin resistance rate = 0.53%) (Carbapenem resistance rate = 23.30%)	1,133			99	90	94			85			86	91	88	84	96	88		83	84	
<i>Acinetobacter baumannii</i> complex (Colistin resistance rate = 1.17%) (Carbapenem resistance rate = 70.48%)	769	40		99	38	46	27		31	31	29	32		31	31	30	30		32	34	
<i>Proteus</i> spp.	361	59			72	96		94	100	81	78	97		99	100	100	100	100	61	62	
<i>Stenotrophomonas maltophilia</i>	323	96	86																	83	
<i>Enterobacter cloacae</i> complex (CRE rate = 7.99%)	313	77			85	97			69	57		61		79	96	96		92	67		
<i>Haemophilus influenzae</i>	183							99			100					100			97	97	79
<i>Haemophilus parainfluenzae</i>	174							99			100					100			64	66	90
<i>Citrobacter</i> spp. (CRE rate = 2.35%)	170	95			95	99		99	89	80		82		95	98	98		98	88		
<i>Morganella morganii</i> (CRE rate = 2.40%)	125	78			89	94			91	83	66	83		96	84	100	99	98	70	70	
<i>Salmonella enterica</i> (intestinal)	101	87																	78		46
<i>Serratia marcescens</i> (CRE rate = 2.02%)	99	94			96	97		96	86	78	93			99	99	99	100	99	89	92	
<i>Klebsiella aerogenes</i> (CRE rate = 2.25%)	89	94			99	100			83	79		78		99	98	100		100	89		
<i>Aeromonas</i> spp.	83	92			98	100			92	83	82	90		96	95	96	96	95	98	97	
<i>Burkholderia cepacia</i> complex	78	100										96				88				82	
<i>Klebsiella oxytoca</i>	45	91			100	98		82	89	91		96		96	100	100		100	87		

0 - 30%

31 - 50%

51 - 70%

71 - 90%

91 - 100%

*The number of each cell represents the number of non-duplicate isolated.

**The number of each cell represents the intermediate percentage in colistin.

Microbiology laboratory, Maharaj Nakorn Chiang Mai hospital

Risk factors of CRE infection



Prior antibiotic exposure



Anti-pseudomonal penicillins

Eg. piperacillin



Beta-lactam/
beta-lactamase inhibitors



Extended-spectrum cephalosporins
(3rd and 4th generation cephalosporin)



Carbapenems



Glycopeptides

Eg. vancomycin, teicoplanin

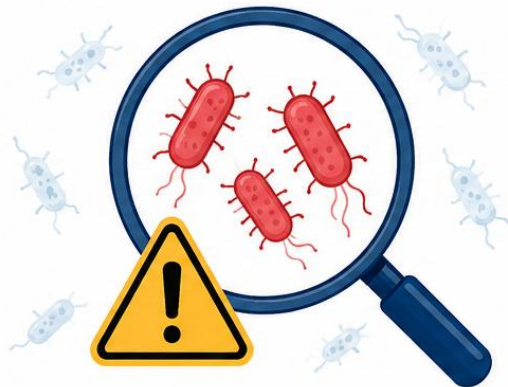


Quinolones

Prior antibiotic exposure



Increased risk of
colonization and/or infection



Patient-related risk factors



Presence of
medical devices



Intensive care unit
(ICU) stay



Solid organ
transplant



Renal failure



Bedbound status

Mechanisms of CRE resistance



1 Not carbapenemase-producing

- Amplification of non-carbapenemase β -lactamase genes (e.g., ESBL genes) with concurrent outer membrane **porin disruption**



2 Carbapenemase-producing



Account for 35%–83% of CRE cases in the United States



Most common carbapenemases in the United States



K. pneumoniae carbapenemases (**KPCs**)

→ not limited to *K. pneumoniae* isolates



New Delhi metallo- β -lactamases (**NDMs**)



Verona integron-encoded metallo- β -lactamases (**VIMs**)



Imipenem-hydrolyzing metallo- β -lactamases (**IMPs**)



Oxacillinases (**OXA-48-like**)

Metallo- β -lactamases (MBLs)



Molecular classifications of **carbapenemase** enzymes

TABLE 1 Molecular classification of carbapenemase enzymes

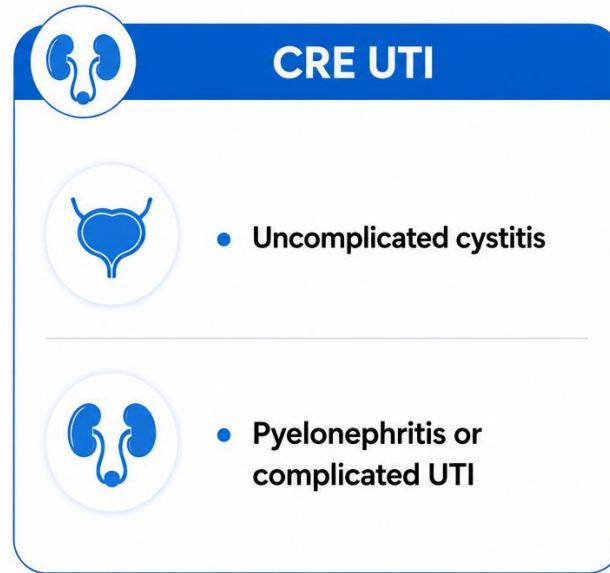
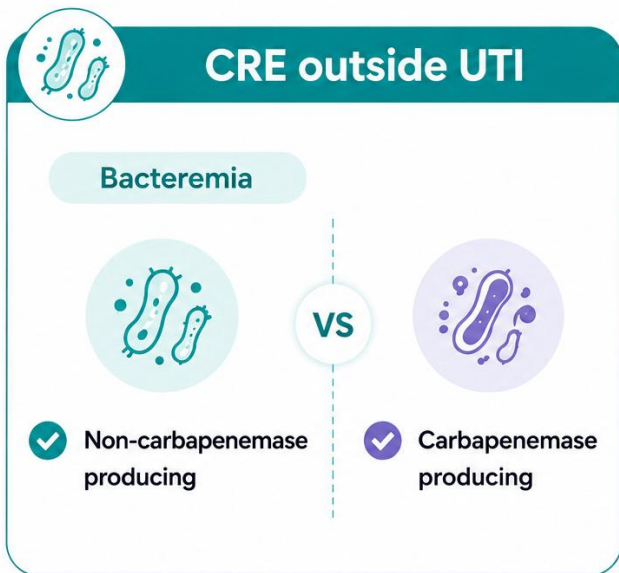
Ambler class	Common enzymes	Microorganisms	Hydrolyzing mechanism	Carbapenem resistance	Aztreonam	Novel active β -lactams	Novel active non- β -lactams
A	KPC, NMC, GES, SME, IMI	<i>K. pneumoniae</i> , <i>E. coli</i> , <i>Serratia marcescens</i> , <i>Enterobacter cloacae</i> , <i>P. aeruginosa</i> , <i>A. baumannii</i>	Serine based	High	Resistant	Ceftazidime-avibactam, ^a aztreonam-avibactam, ^a imipenem-relebactam, meropenem-vaborbactam, cefiderocol	Eravacycline, plazomicin
B	VIM, IMP, NDM (MBLs)	<i>K. pneumoniae</i> , <i>E. coli</i> , <i>S. marcescens</i> , <i>E. cloacae</i> , <i>Citrobacter</i> spp., <i>Proteus</i> spp., <i>Morganella</i> spp., <i>Providencia</i> spp., <i>P. aeruginosa</i>	Zinc based	Variable	Susceptible^b	Aztreonam-avibactam, ^a cefiderocol	Eravacycline, plazomicin ^c
D	OXA (most common, OXA-48-like)	<i>K. pneumoniae</i> , <i>E. coli</i> , <i>Citrobacter</i> spp., <i>Proteus</i> spp.	Serine based	Medium	Susceptible ^b *Some strain nonsusceptible	Ceftazidime-avibactam, aztreonam-avibactam, ^a cefiderocol	Eravacycline, plazomicin

^aCurrently not approved by FDA/EMA.

^bSeveral strains coproduce other β -lactamases (ESBLs, AmpCs, etc.), making them nonsusceptible to aztreonam.

^cNot active against many NDM-producing strains due to the frequent coproduction of 16S rRNA methyltransferase.

Treatment options for CRE infection



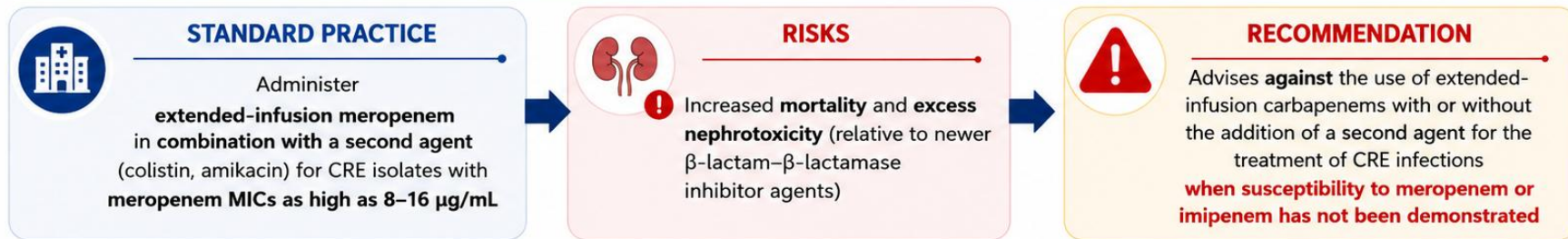
Treatment should be individualized based on site of infection, resistance mechanism, and patient factors.



Treatment of CRE outside UTI

Non-carbapenemase producing

 SUSCEPTIBILITY	 SUGGEST TREATMENT
 Susceptibility to meropenem and imipenem (ie, MICs ≤ 1 $\mu\text{g/mL}$) but are not susceptible to ertapenem (ie, MICs ≥ 1 $\mu\text{g/mL}$)	Use of extended-infusion meropenem (or imipenem–cilastatin) → Infuse over 3 hours 
 Not exhibit susceptibility to any carbapenem	Ceftazidime–avibactam Meropenem–vaborbactam, and Imipenem–cilastatin–relebactam 



Use the most active agent based on susceptibility. De-escalate when possible and monitor toxicity.

Tamma PD, et al. Guidance on the Treatment of Antimicrobial-Resistant Gram-Negative Infections, CID, 2024.



Treatment of CRE outside UTI

Carbapenemase producing

Carbapenemase producing	Preferred treatment	Alternative treatment
 KPC	• Meropenem–vaborbactam • Ceftazidime–avibactam • Imipenem–cilastatin–relebactam ✓ All have >95% activity against KPC-producing <i>Enterobacterales</i> in the US and improved clinical outcomes.	• Cefiderocol • Tigecycline or eravacycline ✓ Tigecycline or eravacycline: for KPC-producing infections not involving the bloodstream or urinary tract. 
 MBL (NDM, VIM, IMP)	• Ceftazidime–avibactam plus aztreonam • Cefiderocol ✓ Active against ~90% of MBL-producing <i>Enterobacterales</i> isolates.	• Tigecycline or eravacycline ✓ For MBL-producing infections not involving the bloodstream or urinary tract. 
 OXA-48	• Ceftazidime–avibactam ✓ >95% of OXA-48-like-producing <i>Enterobacterales</i> isolates are susceptible.	• Cefiderocol • Tigecycline or eravacycline ✓ Tigecycline or eravacycline: for OXA-48-like-producing infections not involving the bloodstream or urinary tract. 



Tailor therapy based on susceptibility, site of infection, patient factors, and local epidemiology.
De-escalate when possible and monitor for efficacy and toxicity.

Tamma PD, et al. *Guidance on the Treatment of Antimicrobial-Resistant Gram-Negative Infections*, CID, 2024.



Treatment of CRE UTI



1 UNCOMPLICATED CYSTITIS caused by CRE

★ PREFERRED TREATMENT



- Nitrofurantoin, TMP-SMX, ciprofloxacin, or levofloxacin, although the likelihood of susceptibility to any of these agents is low
- All achieve **high concentrations in urine**
→ expected to be effective if susceptible

★ ALTERNATIVE TREATMENT



Aminoglycoside (as a single dose)

- Nearly exclusively eliminated by the renal route → single IV dose generally effective for uncomplicated cystitis, with minimal toxicity



Oral fosfomycin (for *E. coli* only)

- *E. coli* resistance to fosfomycin remains rare in the US



Colistin

- Colistin converts to its active form in the urinary tract



- Ceftazidime-avibactam
- Meropenem-vaborbactam
- Imipenemcilastatin-relebactam
- Cefiderocol



2 PYELONEPHRITIS OR COMPLICATED UTI caused by CRE

★ PREFERRED TREATMENT

(clinical trials showing noninferiority to common comparator agents)



TMP-SMX, ciprofloxacin, or levofloxacin (if susceptible)



Ceftazidime-avibactam



Meropenem-vaborbactam



Imipenemcilastatin-relebactam



Cefiderocol

★ ALTERNATIVE TREATMENT



- Aminoglycosides



NOT SUGGEST FOR TREATMENT

- Oral Fosfomycin → limited renal parenchymal concentrations



KEY TAKEAWAY:

Use high-urinary concentration agents for cystitis.

Use systemic agents with proven clinical data for pyelonephritis/complicated UTI.

A 12-year-old male with febrile neutropenia

Management in this patients (2024)

- Diagnosis: CRE bacteremia
- Switch to **ceftazidime/avibactam** 2.5 gm IV drip in 2 hr q 8 hr
- Repeat H/C: persistent GNB



New options?

MBLs ?

- **Ceftazidime/avibactam plus aztreonam IV**
(administered simultaneously via Y-site administration)
- **Aztreonam/avibactam**





Ceftazidime-avibactam?

Aztreonam?

**Y-site
administration?**



Ceftazidime/Avibactam

- Cephalosporins and Betalactamase inh.
- Spectrum: *P. aeruginosa*, Enterobacterales ESBL, KPC, OXA-48 like
- Not cover to MBL Enterobacterales except combined with aztreonam
- Almost excreted by urine

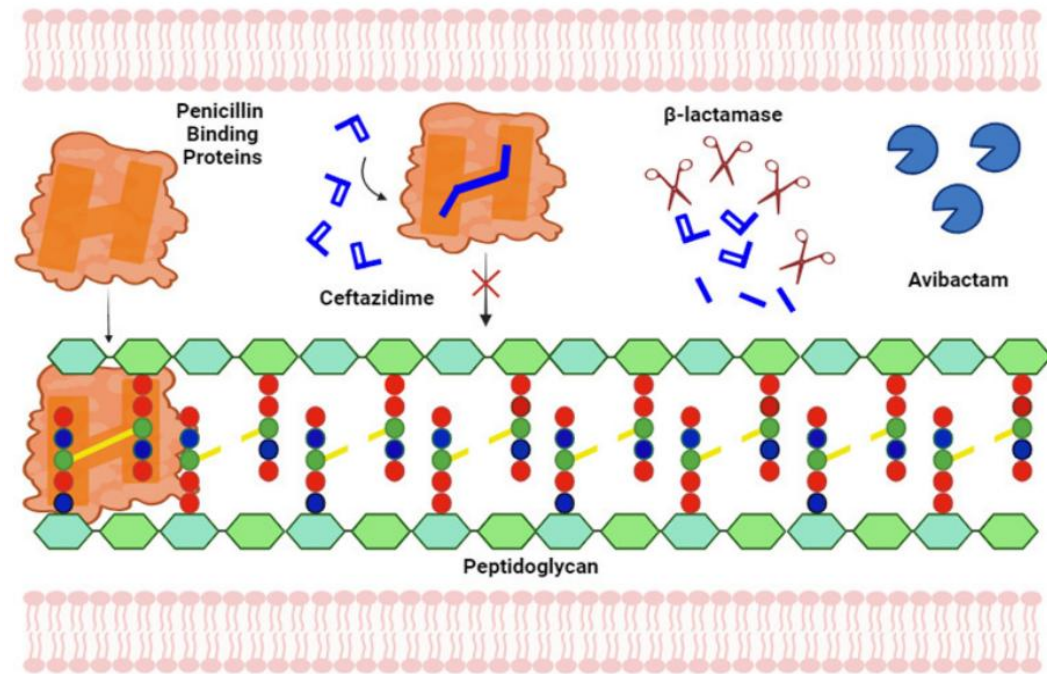


Figure 1. Mechanism of action of ceftazidime/avibactam.

Ceftazidime/Avibactam

- Product: ceftazidime 2g/ avibactam 0.5g inj.
- Dose: (Neonate GA $\geq$ 31 wk, PNA < 29day 25 mg/kg/dose (of ceftazidime) IV q 8 hr *(infusion over 3 hr)
 - 1-3 mo.: 30 mg/kg/dose (of ceftazidime) IV q 8 hr *(infusion over 3 hr)
 - 3-6 mo.: 40 mg/kg/dose (of ceftazidime) IV q 8 hr infusion over 2 hr
 - >6 mo. 50 mg/kg/dose (of ceftazidime) IV q 8 hr max 2 g/dose of ceftazidime infusion over 2 hr
- AE: positive direct comb test, can cause false positive urine sugar test

ศูนย์การศึกษาต่อเนื่องทางเภสัชศาสตร์. Ceftazidime 2 g / Avibactam 0.5 g injection [อินเทอร์เน็ต]. นนทบุรี: สภาเภสัชกรรม; 2567 [เข้าถึงเมื่อ 3 พฤษภาคม 2569].
เข้าถึงได้จาก: https://ccpe.pharmacycouncil.org/index.php?option=article_detail&subpage=article_detail&id=1307

PIDST, Lockowitz CR et al, Journal of the Pediatric Infectious Diseases Society 2025;14(2):1–12

The Use of Ceftazidime–Avibactam in a Pediatric Intensive Care Unit—An Observational Prospective Study

P	37 children (aged 0 to 18 years) admitted to Pediatric Intensive Care Units (PICUs) in Rio de Janeiro, Brazil, with suspected or proven infections due to carbapenem-resistant gram-negative bacteria (CRB).
I	Treatment with Ceftazidime-Avibactam (CAZ-AVI) for more than 24 hours. Median duration of treatment was 11 days.
C	As a descriptive retrospective study, there was no primary control group. However, the study analyzed differences between survivors and non-survivors at 30 days.
O	Clinical/Microbiological: Among 5 confirmed CRB cases, 80% (4/5) achieved microbiological clearance within 7 days. Mortality: The overall 30-day mortality rate was 14/37 (37.8%).

The Use of Ceftazidime–Avibactam in a Pediatric Intensive Care Unit—An Observational Prospective Study

- 37 children, mean age 28 months (range 1-215 months)
- Reasons of prescription CTZ/AVI
 - Lower respiratory tract infection: 17/37(45.9%)
 - Sepsis or septic shock: 6/37(16.2%)
 - Two or more sites of infection 4/37(10.8%)
 - GI tract infection: 2/37(5.4%)
 - CNS infection: 2/37(5.4%)
 - Other reasons: 4/37(10.8%)

Aztreonam

- Monobactam, against gram negative bacteria via inh. cell wall synthesis by bind to PBPs (PBP3)
- Not hydrolysed by MBL (NDM, IMP, VIM) producing Enterobacterales enz.
- Not cover to Gram positive, anaerobes
- Hydrolysed by OXA, KPC, ESBL, AmpC producing Enterobacterales

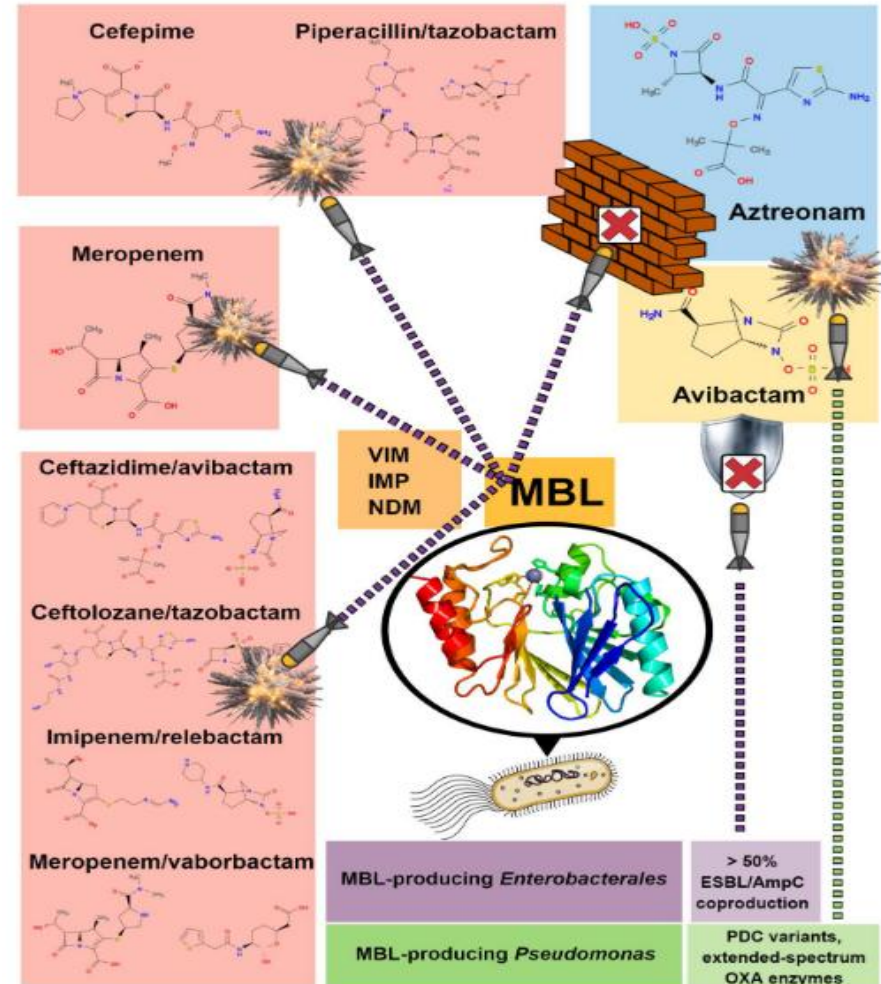


Figure 1. β -lactam antibiotic targets of MBL enzymatic activity.

Aztreonam

- For CRE with NDM or MBL production
 - **Aztreonam** + Ceftazidime / Avibactam
- Dose infant, children, adult: 90-120 mg/kg/day divided in q 6-8 hr
max 8 g/day (**infusion >3hr**)

• Neonate	Gestation	Post natal age	IV Dose
	< 34 weeks gestation	< 7 days	30 mg/kg/dose given 12 hourly
		≥7 days	30 mg/kg/dose given 8 hourly
	≥ 34 weeks gestation	< 7 days	30 mg/kg/dose given 8 hourly
		≥7 days	30 mg/kg/dose given 6 hourly

Y-site incompatibility

- Acyclovir
- Amphotericin B
 - deoxycolate amphotericin B
 - liposomal amphotericin B
- Diazepam
- Ganciclovir
- Metronidazole
- Pantoprazole
- Pentamidine
- Phenytoin

Efficacy of Ceftazidime-avibactam Plus Aztreonam in Patients With Bloodstream Infections Caused by Metallo- β -lactamase–Producing Enterobacterales

P (Population)	102 adult patients (≥ 18 years) with bloodstream infections (BSIs) caused by metallo-β-lactamase (MBL)-producing Enterobacterales (NDM and VIM strains).
I (Intervention)	Targeted therapy with a combination of Ceftazidime-avibactam (CAZ-AVI) (2.5 g every 8h) and Aztreonam (ATM) (2 g every 8h).
C (Comparison)	Targeted therapy with other active antibiotics (OAAs) , such as colistin-based regimens, or other active agents like tigecycline and fosfomycin.
O (Outcome)	Primary: 30-day all-cause mortality. Secondary: Clinical failure at day 14 and length of hospital stay (LOS).

Table 2.
Targeted
Antibiotic
Regimens
Administered in
102 Bloodstream
Infections Due to
MBLProducing
Enterobacterales

Antibiotic Regimen	No. (%) (N = 102)	Mortality, No. (%)
CAZ-AVI + Aztreonam (ATM)	52 (51)	10/52 (19.2)
Others active antimicrobials		
Colistin-containing regimens	27 (26.5)	16/27 (59.3)
* Colistin + fosfomycin + tigecycline	7	6/7
* Colistin + fosfomycin	7	5/7
* Colistin + meropenem	5	3/5
* Colistin + ATM ± piperacillin-tazobactam	4	1/4
* Colistin + gentamicin	1	0/1
* Colistin + cotrimoxazole	1	0/1
* Colistin alone	2	1/2
Regimens not containing colistin	23 (22.5)	6/23 (26.1)
* Tigecycline + aminoglycosides	8	2/8
* Fosfomycin + aminoglycosides	5	0/5
* Tigecycline + fosfomycin	2	2/2
* Tigecycline + meropenem	1	0/1
* ATM + aminoglycosides	4	1/4
* ATM + fosfomycin	1	0/1
* ATM alone	2	1/2

Outcome Measure	CAZ-AVI + ATM (n=52)	Other Active Antibiotics (OAAs) (n=50)	P-Value
30-Day Mortality	19.2%	44%	.007
Clinical Failure (Day 14)	25%	52%	.005
Drug-Induced Kidney Injury	1.9%	20%	.003

Outcome Category	Key Findings
Mortality Reduction	The CAZ-AVI + ATM group had a 60% to 70% lower risk of death (after adjusting for patient health and severity via propensity score matching).
Speed of Recovery	Patients receiving the combination therapy had a significantly shorter length of hospital stay after the onset of the infection.
Safety	The combination was much less toxic to the kidneys than colistin-based regimens, which are often the "standard" alternative for these resistant infections.

Case summary



Case study 1: CRAB VAP

- Increasing resistance of CRAB
- Antibiotics options: sulbactam-based regimen, polymyxin B (less nephrotoxicity)



Sulbactam

Key component of sulbactam-based regimens



Polymyxin B

Consider in patients with renal concerns (less nephrotoxicity)



Case study 2: CRE bacteremia

- High mortality especially in immunocompromised host
- Novel antibiotics: ceftazidime/avibactam, aztreonam/avibactam



Ceftazidime/ avibactam

Active against many CRE organisms



Aztreonam

Useful option especially in combination strategies



Antibiotic stewardship is still essential to control AMR



Right drug



Right dose



Right duration



Monitor &
de-escalation



Thank you

· Q&A ·

Together we can combat AMR



Protect today. Preserve tomorrow.



Prevent

Infection prevention
saves lives



Detect

Timely diagnosis
guides effective care



Treat

Appropriate antibiotics
for better outcomes



Together

Stewardship today,
healthier future



Every decision counts. **Every** action matters. **Every** child deserves a future.

Guidance on antibiotic therapy in children with drug-resistant Gram-negative infections

- ## 2. คำจำกัดความ

- Multidrug-Resistant Organisms (MDROs) អង្គរ

เชื้อก่อโรคโดยเฉพาะเชื้อแบคทีเรียที่ติดต่อจากปัสสาวะที่ใช้ในการรักษาล้างแค้น 3 กลุ่มขึ้นไป

Extended-spectrum beta-lactamases producing Enterobacteriales (ESBLs) หมายถึงเชื้อจาก Enterobacteriales ที่สามารถสร้างเอนไซม์ extended-spectrum beta-lactamases โดยเอนไซม์นี้จะทำลายยาปฏิชีวนะในกลุ่ม penicillins และ third generation-cephalosporins แต่ถูกยับยั้งได้ด้วย beta-lactamase inhibitors พิจารณาจากผลความไวของเชื้อต่อยาปฏิชีวนะส่วนใหญ่พบว่าเชื้อที่ดื้อยา cefotaxime, cefepime แต่ไม่ดื้อกับยาไวต่อยา ceftazidime, amoxicillin-clavulanic acid, piperacillin-tazobactam, carbapenems

- Plasmid-mediated AmpC beta-lactamases producing Enterobacterales (AmpC)

พลาสมิด เชื้อกลุ่ม Enterobacterales ที่สามารถสร้างเอนไซม์ AmpC beta-lactamase โดยเอนไซม์นี้จะทำลายยาปฏิชีวนะกลุ่ม penicillins และ third generation-cephalosporins และไม่สามารถยับยั้งได้ด้วย beta-lactamase inhibitors พิจารณาจากผลความไวของเชื้อต่อยาปฏิชีวนะส่วนใหญ่พบว่าเชื้อจะดื้อต่อยา cefotixin, ceftriaxone, amoxicillin-clavulanic acid, piperacillin-tazobactam แต่จะยังพอมีความไวต่อยา cefepime และ amikacin-carbapenems

หมายเหตุ ในแนวทางนี้ จะขอจัดกลุ่ม ESBLs และ AmpC เป็นกลุ่ม third-generation cephalosporins-resistant Enterobacterales เนื่องจากในทางปฏิบัติ ห้องปฏิบัติการไม่ได้ตรวจการสร้างเอนไซม์ดังกล่าว

- Carbapenem-resistant *Enterobacterales* (CRE) หมายถึง เชื้อกลุ่ม *Enterobacterales* ที่ดื้อต่อยาต้าน carapenems อย่างน้อย 1 ชนิด ได้แก่ ertapenem, meropenem, imipenem/cilastatin, doripenem

- Carbapenem-resistant *Acinetobacter baumannii* (CRAB) หมายถึง เชื้อ *A. baumannii* ที่ดื้อต่อกับกลุ่ม carbapenems เช่น meropenem, imipenem/cilastatin, doripenem ยกเว้นยา ertapenem ซึ่งไม่มีฤทธิ์ครอบคลุมเชื้อ *A. baumannii* CRAB ในทั้งหมดยา รวมถึงเชื้อ *Acinetobacter* สายพันธุ์อื่น ๆ ในกลุ่ม *baumannii* และ *calcoaceticus* complexes

Pranita D. Tamma,^{1,6} Emily L. Neil,² Julie Ann Justo,³ Amy J. Mathers,⁴ Michael J. Satlin,⁵ and Robert A. Bonomo¹

The Infectious Diseases Society of America (IDSA) is committed to providing up-to-date guidance on the treatment of

The Infectious Diseases Society of America (IDSA) is committed to providing up-to-date guidance on the treatment of antimicrobial-resistant (AMR) infections. This guidance document focuses on infections caused by extended spectrum beta-lactamase (ESBL)-producing *Enterobacteriaceae*, including *Escherichia coli* (E. coli), *Klebsiella pneumoniae* (K. pneumoniae), and *Pseudomonas aeruginosa* (P. aeruginosa), carbapenem-resistant *Enterobacteriaceae* (CRE), *Pseudomonas aeruginosa* with difficult-to-treat resistance (DTRP *P. aeruginosa*), carbapenem-resistant *Acinetobacter baumannii* (CRAB), and *Stenotrophomonas maltophilia* (S. maltophilia). This document is a revised version of the guidance document, “A panel of 6 infectious diseases specialists” with expertise in managing antimicrobial-resistant infections formulated questions about the treatment of infections caused by ESBL-E, Amp-C-E, CRE, DTRP *P. aeruginosa*, CRAB, and S. maltophilia. The document is intended to be used by clinicians and researchers, and is not intended to be used for clinical decision-making. The document is being updated and revised infrequently; therefore, this document focuses on the treatment of AMR infections in the United States. Preferred and alternative suggested treatment approaches are provided with accompanying rationale, assuming the causative organism is confirmed by culture and sensitivity testing. The document also includes information on the duration of therapy, duration of therapy, and other management considerations by discussing rapidly. Suggested approaches apply for both adult and pediatric patients, and the document is intended to be used by clinicians and researchers. The document is intended to be used by clinicians and researchers. Consultation with an infectious diseases specialist is recommended for the treatment of AMR infections. This document is current as of December 31, 2023 and will be updated periodically. The most current version of this document, including data

Keywords. ESBL; *Pseudomonas aeruginosa*; CRAB; *Stenotrophomonas maltophilia*.

Antimicrobial-resistant (AMR) infections are a global crisis. Internationally, approximately 1.3 million deaths were estimated to be directly attributable to AMR pathogens in 2019 [1]. In the United States, AMR pathogens caused more than 2.8 million infections and over 35 000 deaths annually from 2012 through 2017, according to the Centers for Disease Control and Prevention (CDC) Antibiotic Resistance Threats in the United States Report [2].

As an alternative to practice guidelines, the Infectious Diseases Society of America (IDSA) has endorsed developing more narrowly focused guidance documents for the treatment of infections where data may not be very robust and continue to rapidly evolve – such as with AMR. Guidance documents are prepared by a small team of experts, who answer questions about treatment based on a comprehensive (but not necessarily systematic) review of the literature, clinical experience, and expert opinion. Documents are made available online and updated annually.

In the present document, guidance is provided on the treatment of infections caused by extended-spectrum β -lactamase-producing Enterobacteriales (ESBL-E). AmpC β -lactamase-producing Enterobacteriales (AmpC-E), carbapenem-resistant Enterobacteriales (CRE), *Pseudomonas aeruginosa* with difficult-to-treat resistance (DTTR; *P. aeruginosa*), carbapenem-resistant *Acinetobacter baumannii* (CRAB), and *Stenotrophomonas maltophilia*. Many of these pathogens have been designated urgent or serious threats by the CDC [2]. Each pathogen causes a wide range of infections that are encountered in United States hospitals of

AMR Treatment Guidance • CID • 1

VIEW

Christine R. Lockowitz¹, Alice J. Hour², Kathleen Chiotos³, Laura L. Bir⁴, Aimee M. Dassner⁵, Andrew B. Gainey⁶, Jennifer E. Ginotra⁷, Denise Iacono⁸, Taylor Mericciotti^{9,10}, Grant Stines¹¹, M. Tami Tsur¹², William S. Wilson¹³, Pranita D. Thomas¹⁴

[illegible]

The Infectious Diseases Society of America (IDSA) publishes annual guidance on the treatment of antimicrobial resistance (AMR) gram-negative infections. Within the AMR guidance, suggested dosages of antibiotics for adults infected with AMR pathogens are provided. This document serves as a companion document to the IDSA guidance to assist pediatric specialists with dosing 13 pediatric agents for the treatment of AMR infections in children. A panel of 13 pediatric infectious diseases specialists, including 11 pharmacists and 2 physicians, reviewed the pharmacokinetic/pharmacodynamic (PK/PD) and clinical data for more than 1,000 children with AMR infections in the real world. The panel identified 13 pediatric agents for AMR infections: ampicillin, ampicillin-sulbactam, ceftazidime-avibactam and avibactam, ceftazidime-tazobactam, imipenem cilastatin, cefiderocol, meropenem, vibactam, sulbactam-doripenem, and doripenem. Suggested dosing for ampicillin-sulbactam is also provided, given complexities in dosing for carbapenem resistance. *Acinetobacter baumannii* infections, *Candida*-substant suggested dosing for 13-agent agents used to treat AMR infections in neonates, infants, and adolescents and relevant supporting evidence are provided. Critical to us up to date as of December 2023, the panel also reviewed the existing data on optimizing antibiotic dosing to improve the outcomes of children with AMR infections.

Key words: ampicillin-sulbactam; cefiderocol; ceftaridibactam; meropenem-vaborbactam; sulbactam-durlobactam.

INTRODUCTION

The Infectious Diseases Society of America (IDSA) publishes annual guidance on the treatment of antimicrobial-resistant (AMR) infections, including infections caused by extended-spectrum β -lactamase-producing *Enterobacteriales* (ESBL-E), AmpC β -lactamase-producing *Enterobacteriales* (AmpC-E), carbapenem-resistant *Enterobacteriales* (CRP-

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<https://doi.org/10.1093/pids/pia064>

Parasitoma aeruginosa with difficult-to-treat resistance (DTT *P. aeruginosa*), carbapenem-resistant *Acinetobacter baumannii* (CRAB), and *Streptophomonas multiphila*).¹ The IDSA guidance document includes the rationale for suggested management approaches, which can be applied to both adults and children; however, it only provides antibiotic dosing for adults with normal renal and hepatic function. The current document is meant to serve as a companion to the IDSA guidance to assist clinicians with suggested dosing for novel β -lactam antibiotics used to treat AMR infections in children (ie, ceftazidime-avibactam, ceftazidime-avibactam and atezizumab and ceftolozane-tazobactam, imipenem-cilastatin-relbactam, meropenem-vaibactam, sulbactam-durlobactam). Suggestions for ampicillin-sulbactam is also provided, given the complexities in appropriately dosing this agent for CRAB.

Journal of the Pediatric Infectious Diseases Society, 2025, Vol. 14, No. 2 • JPHIDS 2025;14 (2) •

Thailand One Health Dashboard on Antimicrobial Resistance

Data on monitoring and evaluation of AMR policies

Information on this and other topics is available online at <http://www.pearsoned.com>

Antimicrobial Consumption
million DDDs

Antimicrobial Resistance



Knowledge and Awareness on AMI

<https://www.thaiamrwatch.net/>